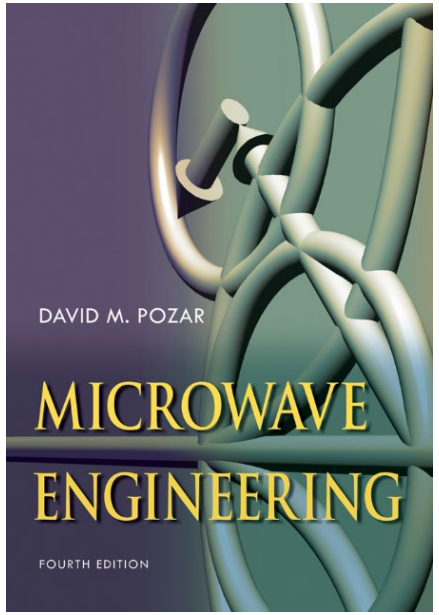
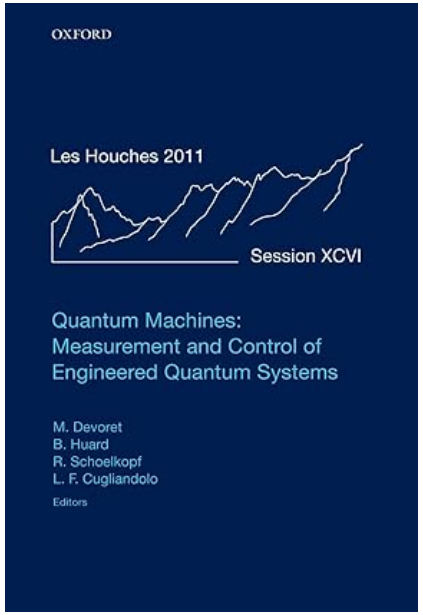
Radio Frequency Electronics for Quantum Thechnologies



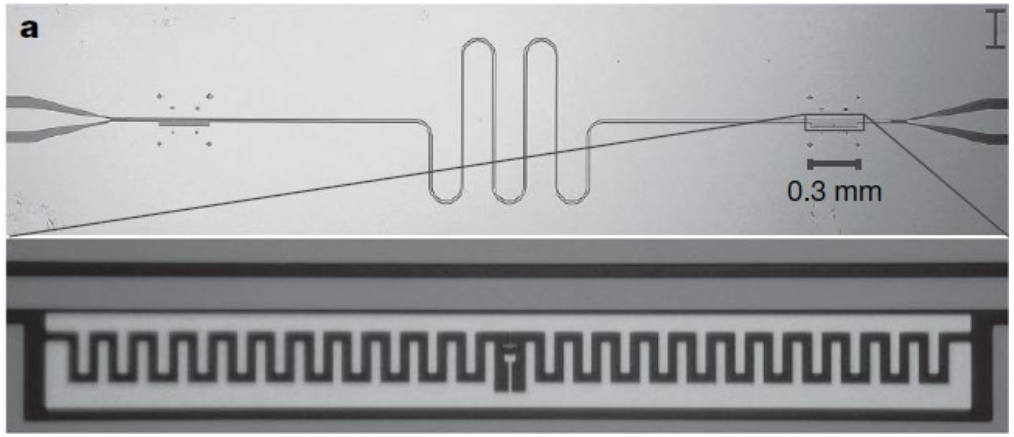
SONNET®
PRECISION ELECTROMAGNETICS

scikit-rf
Object Oriented RF Engineering

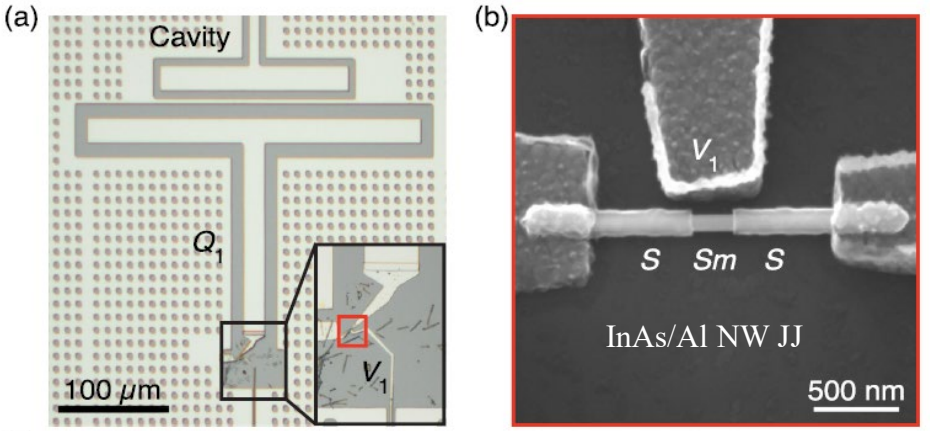


Radio frequency electronics for quantum thechnologies

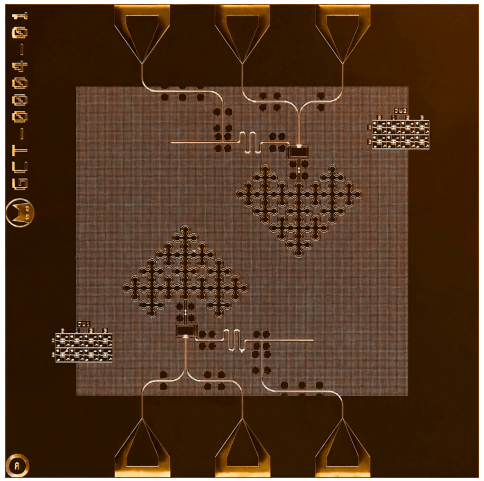
Circuits that we would like to design / simulate:



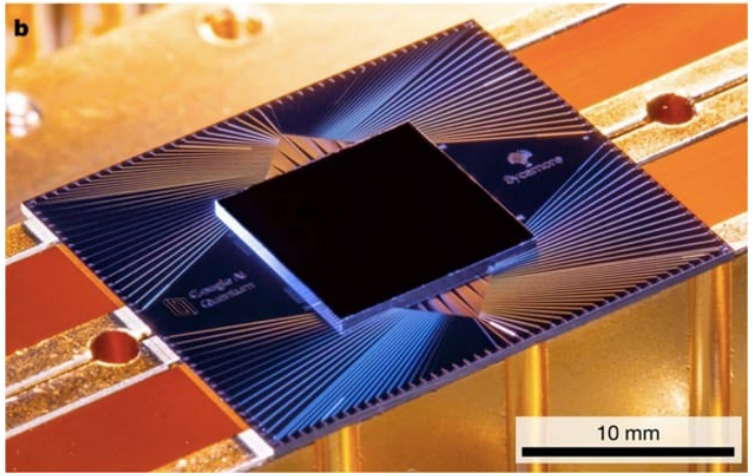
Nature **454**, 315–318 (2008), Wallraff's group



PRB **97**, 060508(R) (2018) Marcus group



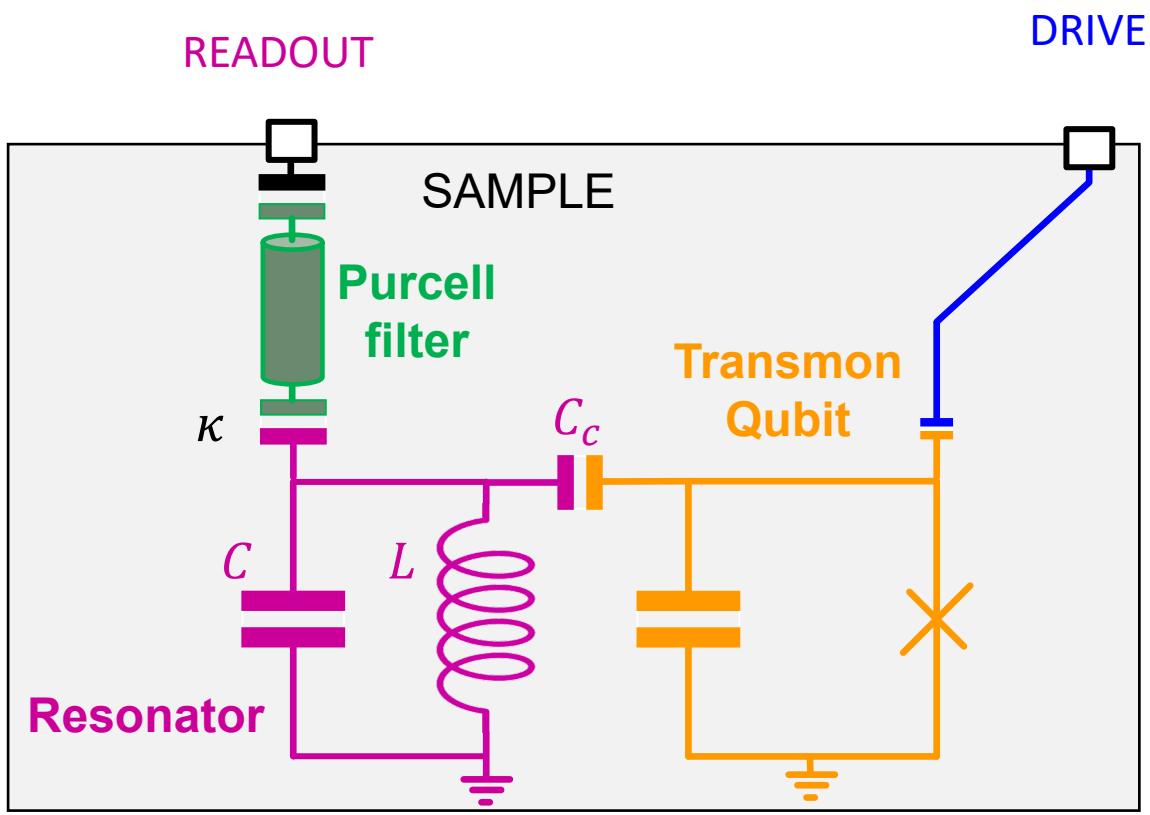
Boson 4 board, Alice & Bob (2024)

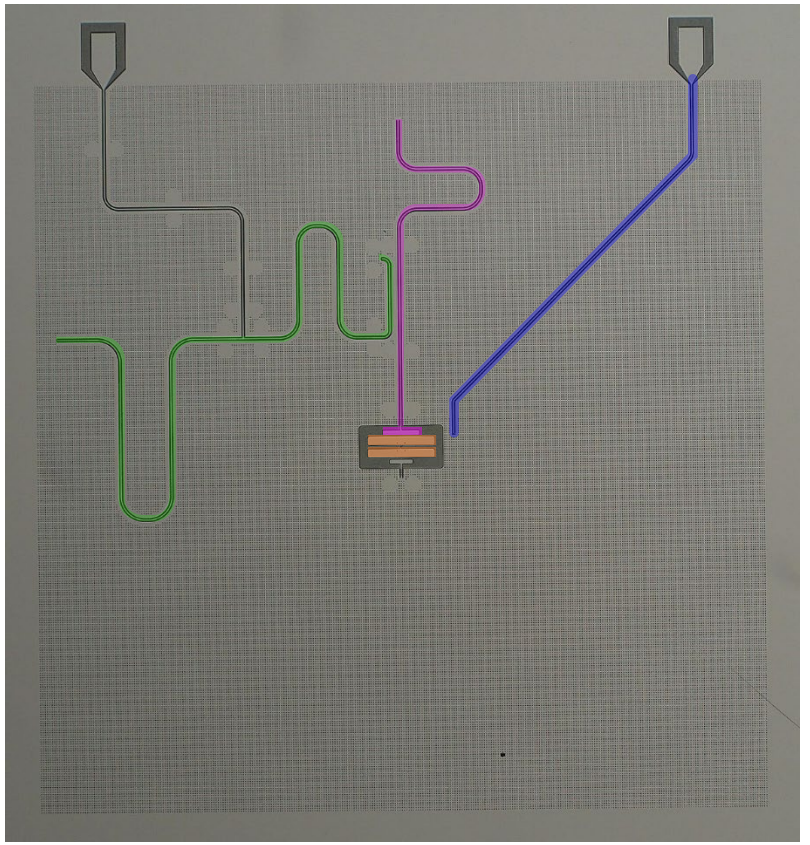


Sycamore processor, Google (53 qubits)

Dispersive Measurement of a Transmon Qubit

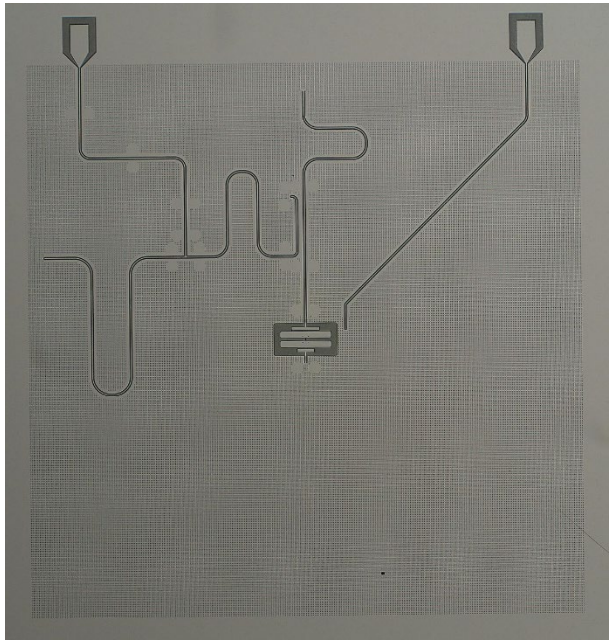
From Macroscopic Quantum Tunneling to Circuit QED





Outline

- Quantum Macroscopic Tunneling
- Josephson Junction as a Quantum Nonlinear Oscillator
- The Transmon Qubit
  - Simulation of the capacitance
  - Calculation of the equivalent inductance
- Coupling the Transmon to a Microwave Resonator
  - Simulation of a quarter wave resonator
  - Simulation of the coupling
- The Dispersive Regime
- Principle of Dispersive Measurement
- Simulation of the whole circuit

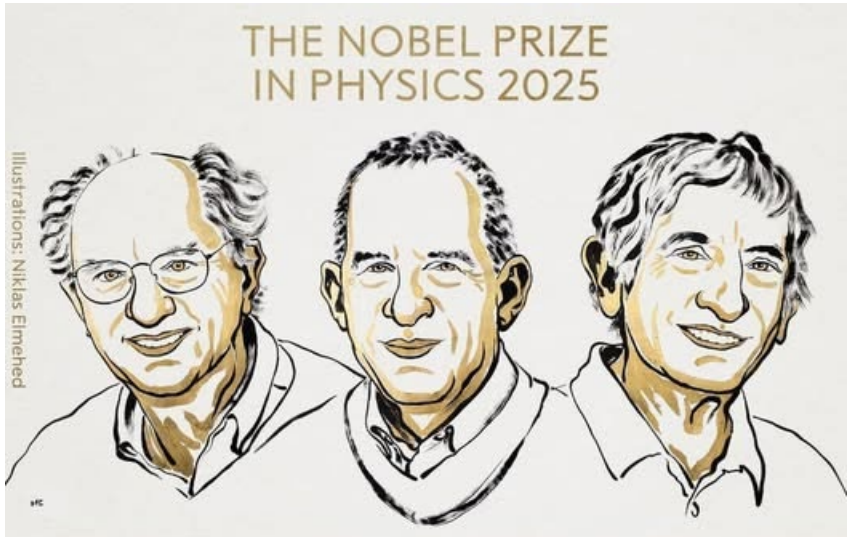


Quantum Macroscopic Tunneling

The Nobel Prize in Physics 2025

" for the discovery of macroscopic quantum mechanical tunnelling and energy quantization in an electric circuit "

"This brings quantum physics from the subatomic world onto a chip" (Göran Johansson)



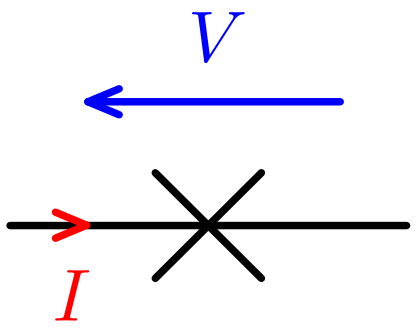
- The tunneling effect was well known for microscopic object
  - Double potential wells for studies of molecular spectra [F. Hund, Z. Phys. 40, 742 (1927)]
  - Field emission of electrons from a metal surface by tunneling [R. H. Fowler, L. Nordheim, Proc. R. Soc. London (1928))
  - Tunneling of alpha particles of radioactive decay [G. Gamow, Z. Phys. 51, 204 (1928)]
- **Quantum Macroscopic Tunneling (QMT)** refers to the phenomenon by which a **macroscopic object (or a collective variable)** traverses a potential barrier **without sufficient energy to overcome it classically**, thanks to quantum properties

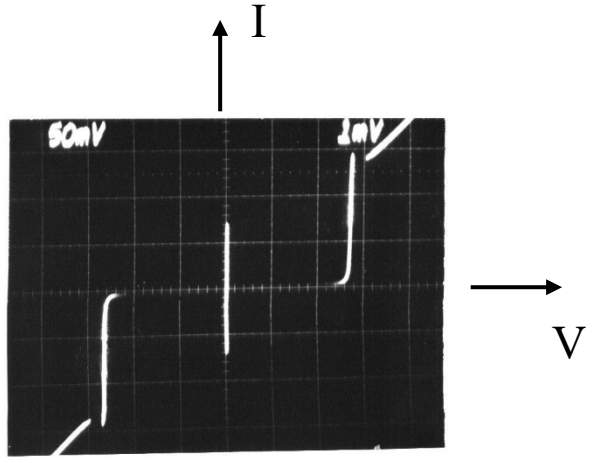
Josephson Junction as a Macroscopic Quantum System

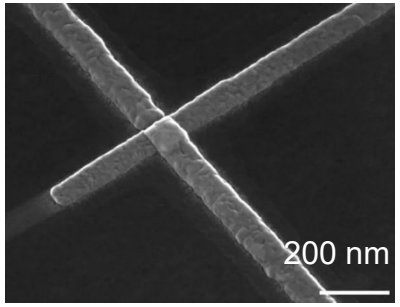
- In macroscopic systems, Quantum Macroscopic Tunneling (QMT) does not involve an individual particle but rather a collective variable that describes the state of the entire system.
- Josephson junctions are paradigmatic examples of macroscopic systems exhibiting quantum effects. They consist of two superconductors separated by a thin insulating layer (or a weak link) [B. Josephson Phys. Rev, 1 7 251 (1967)]
- The Josephson effect is the quantum tunneling of superconducting Cooper pairs through a thin barrier, enabling current flow without applied voltage.
- JJ is a is **lossless and strongly nonlinear**, it behaves as a **non-linear inductor**,

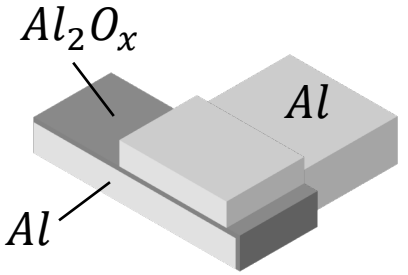
$$I = I_c \sin\varphi$$

$$V = \frac{\hbar}{2e}\frac{d\varphi}{dt}$$

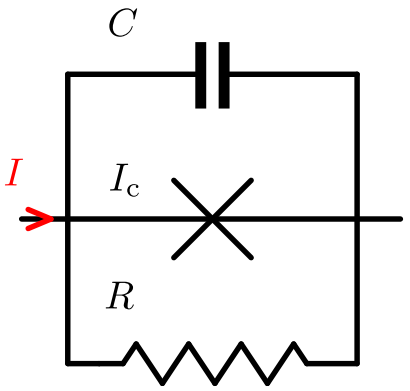








Macroscopic Quantum Tunneling



– RCSJ model (Resistively and Capacitively Shunted Junction)

classical dynamics of the superconducting phase:

$$C\left(\frac{\hbar}{2e}\right)^2\ddot{\varphi}+\frac{1}{R}\left(\frac{\hbar}{2e}\right)^2\dot{\varphi}+\frac{\partial U}{\partial\varphi}=0 \quad \text{with} \quad U(\varphi)=-E_{\mathrm{J}}\left(\cos\varphi+\frac{I}{I_{\mathrm{c}}}\varphi\right) \quad \text{and} \quad E_{\mathrm{J}}=\frac{\hbar}{2e}I_{\mathrm{c}}$$

– MQT Escape rate (instanton method)

$$\omega_{\mathrm{p}}(I)=\sqrt{\frac{2eI_{\mathrm{c}}}{\hbar C}}\left(1-\left(\frac{I}{I_{\mathrm{c}}}\right)^2\right)^{1/4}$$

$$\Gamma_{\mathrm{MQT}}=\frac{\omega_{\mathrm{p}}}{2\pi}\sqrt{120\pi\frac{\Delta U}{\hbar\omega_{\mathrm{p}}}}\exp\left(-\frac{36}{5}\frac{\Delta U}{\hbar\omega_{\mathrm{p}}}\right)$$

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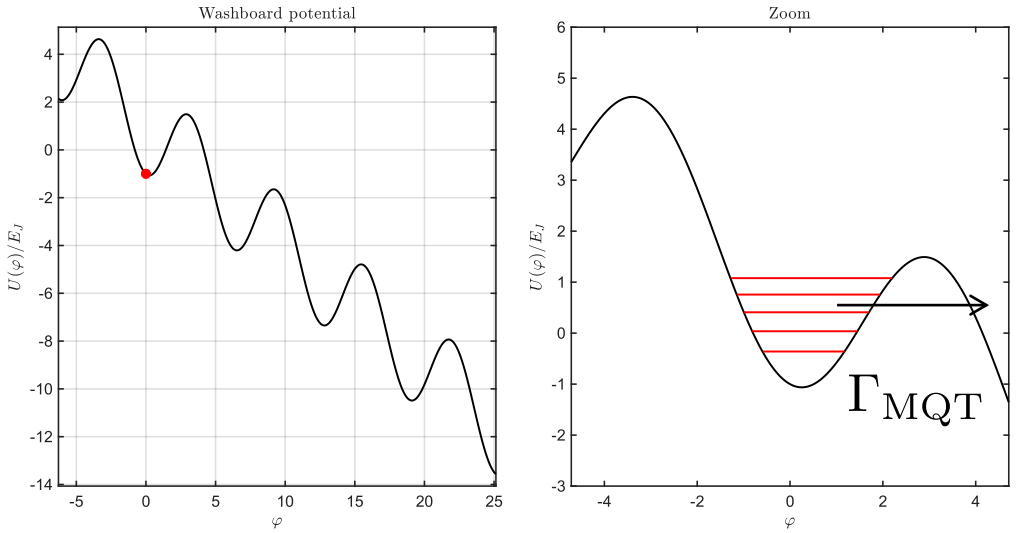
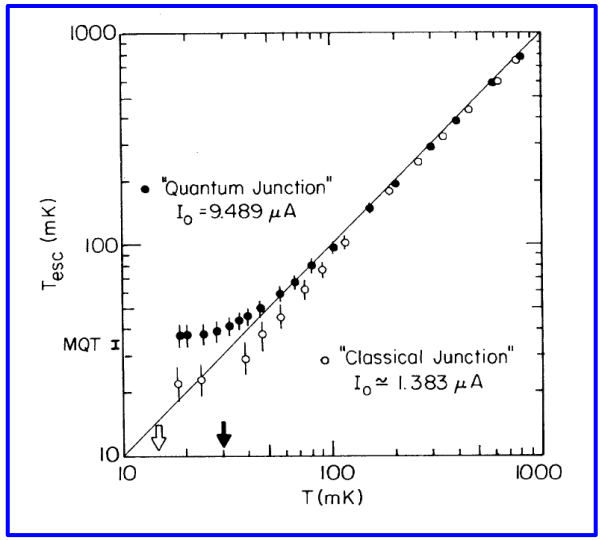
Josephson Junction as an artificial atom

– The macroscopic quantum tunneling and the first artificial atom

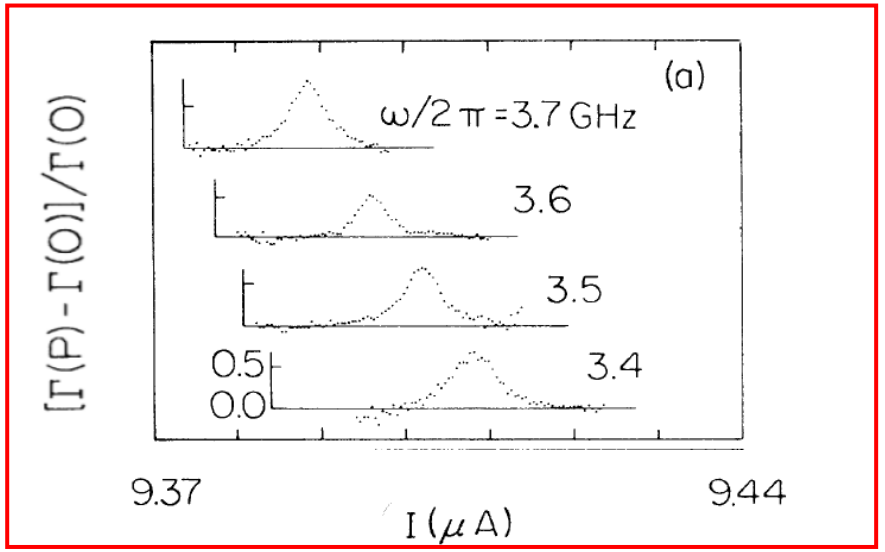


$n_e\xi^3 \sim 10^{10}$ particles



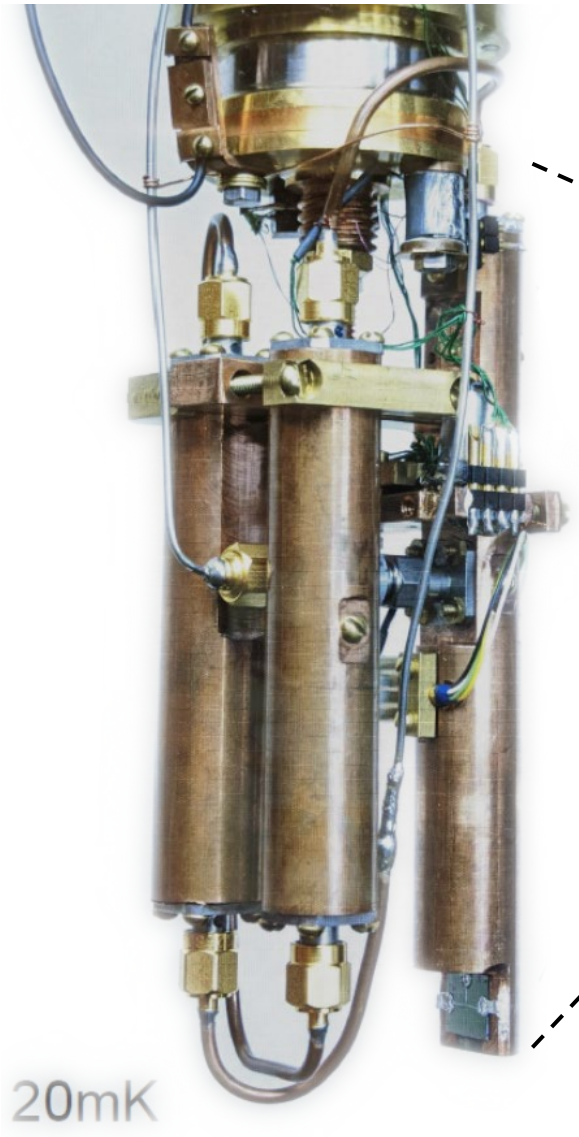
PhysRevLett.55.1908 (1985)

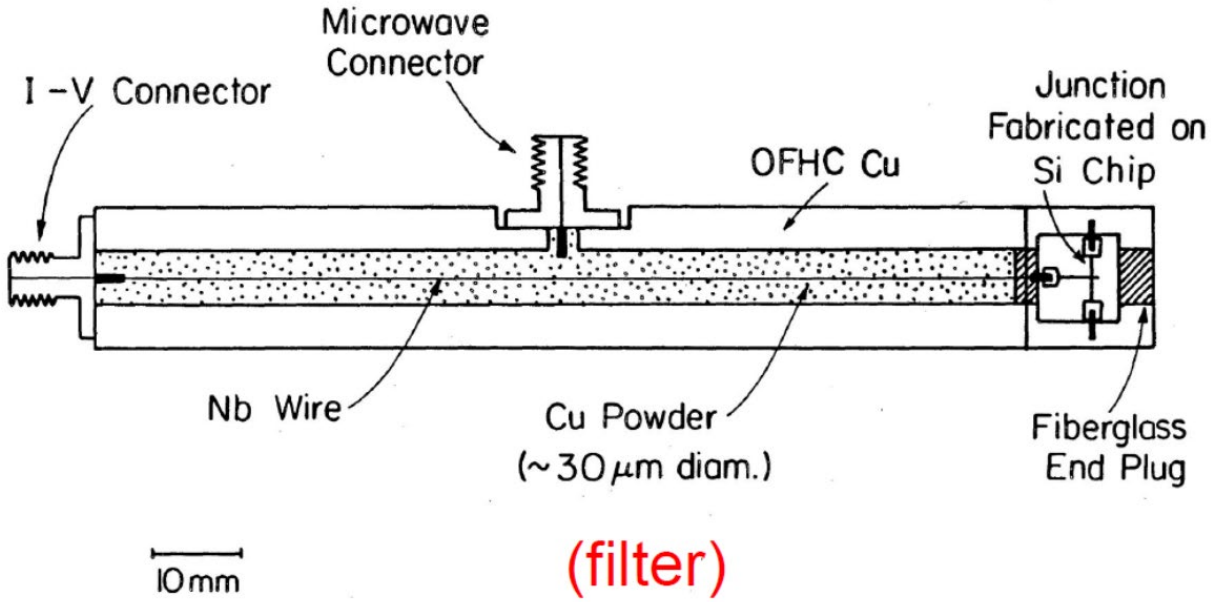
– These results provide very strong evidence for the existence of energy quantization and macroscopic quantum mechanical tunneling in an electrical circuit



PhysRevLett.55.1543 (1985)

Experimental setup





Clark, Devoret, Martinis experiment

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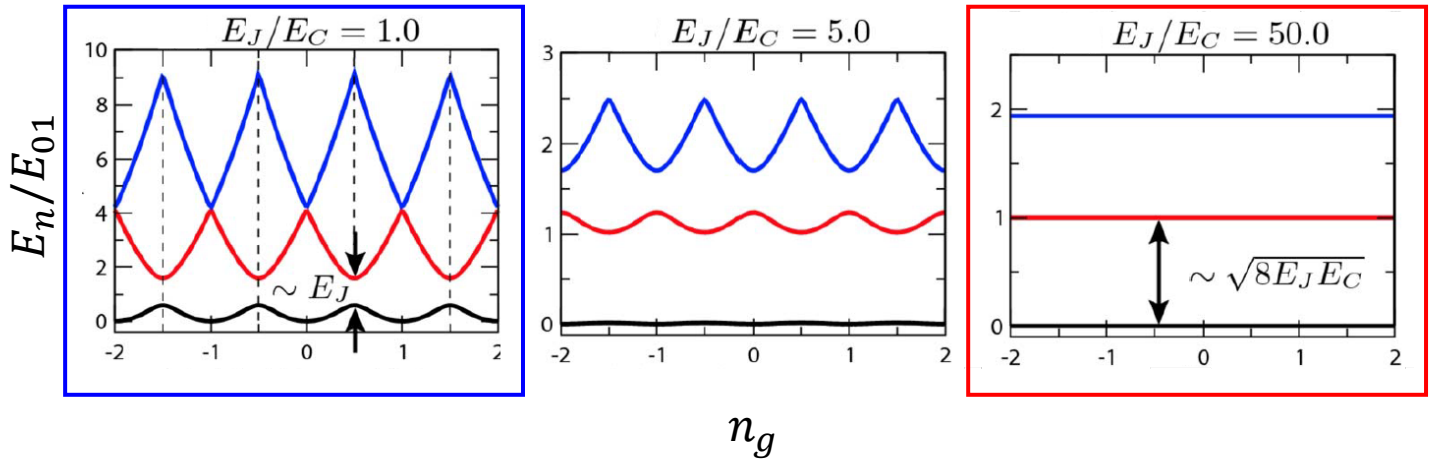
The Transmon Qubit

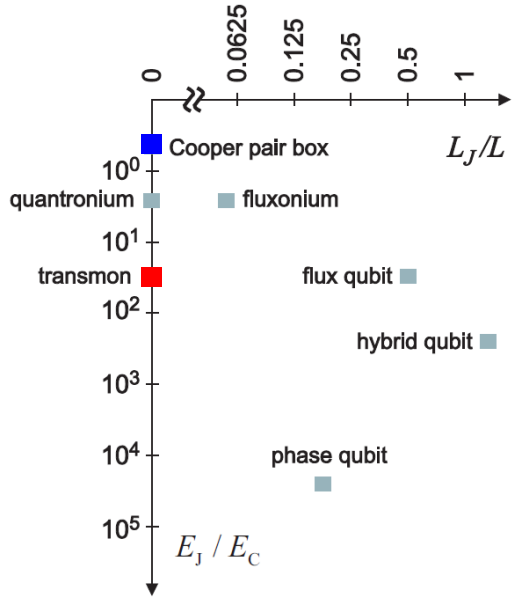
- The superconducting qubit is a lossless, nonlinear LC circuit. Dynamical variables—the node flux ϕ and the conjugate charge Q are treated as quantum degrees of freedom such that $[\hat{\phi}, \hat{Q}] = i\hbar$.

- The Hamiltonian of a superconducting qubits is $H = 4E_c(\hat{n} - n_g)^2 - E_J \cos\hat{\phi}$, with $E_c = \frac{e^2}{2C}$ and $E_J = \frac{\hbar I_c}{2e} = \frac{R_k}{8R_n}\Delta$; $\hat{Q} = 2e\hat{n}$ and n_g is the offset charge

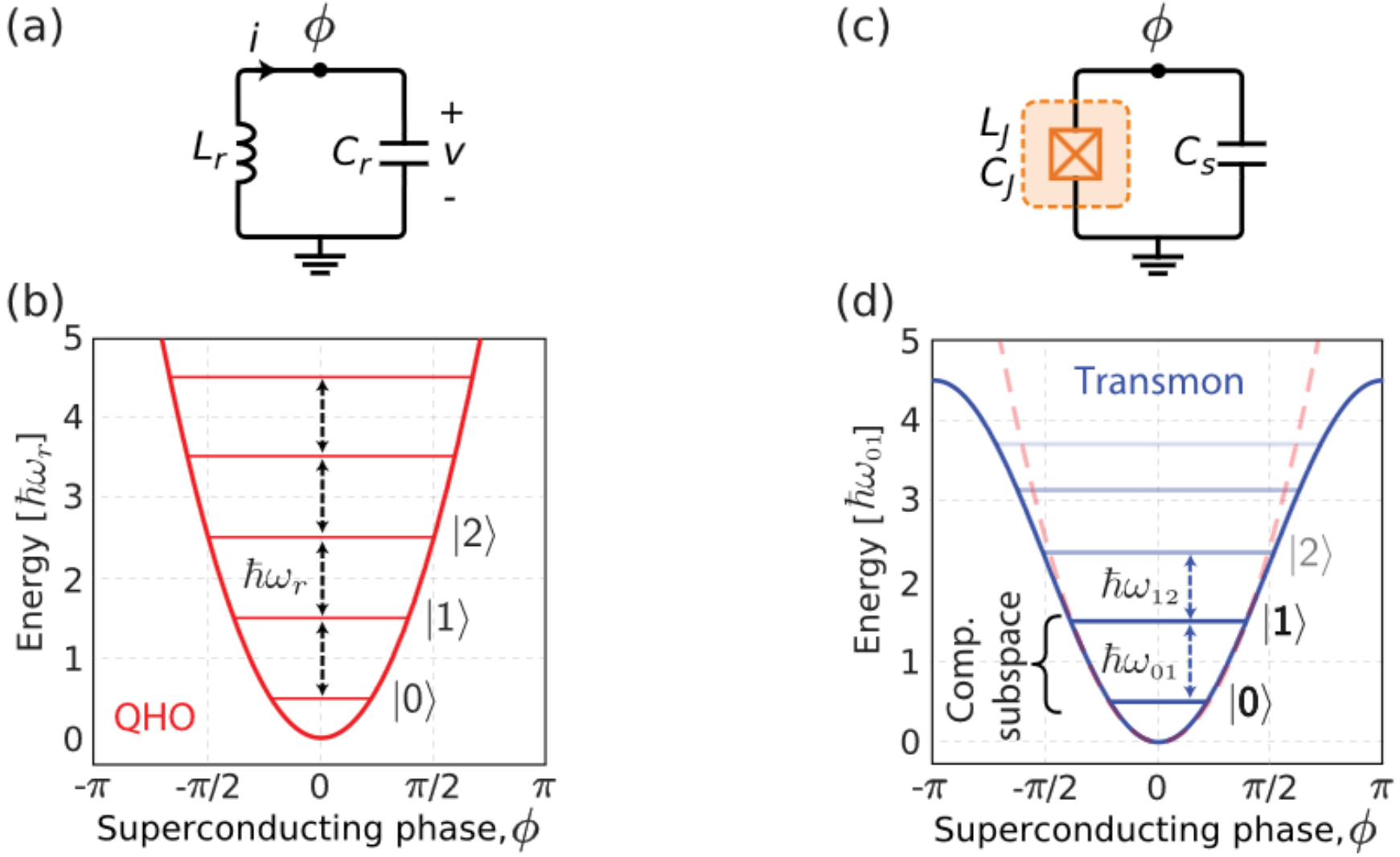
- In the usual transmon regime $E_J \gg E_c$, we expand the cosine to get a weakly anharmonic oscillator:

$$H = 4E_c(\hat{n} - n_g)^2 + \frac{1}{2}E_J\hat{\phi}^2 - \frac{1}{24}E_J\hat{\phi}^4$$





The Transmon Qubit

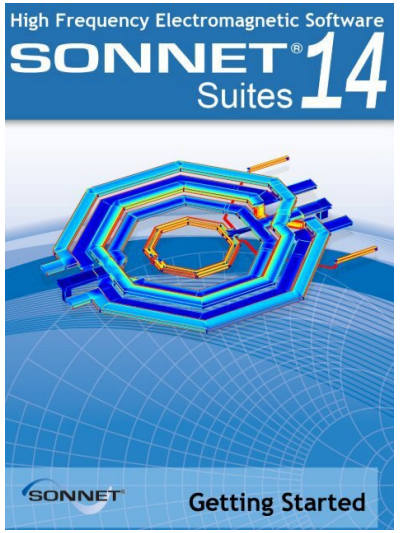


https://jupyter.lps.u-psud.fr

Intermezzo : simulation with Sonnet

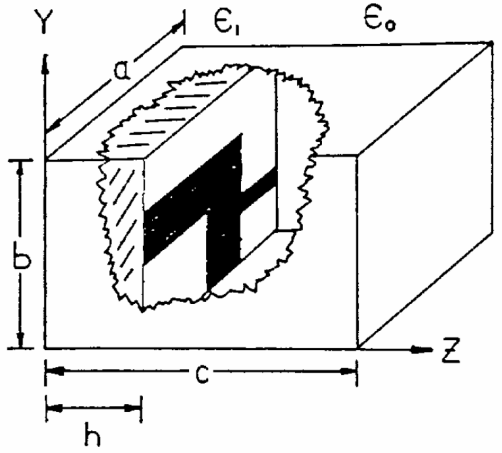
Overview of the Sonnet 3D Planar EM Simulator

Sonnet's suites of high-frequency electromagnetic (EM) Software are aimed at design involving predominantly planar (3D planar) circuits and antennas. Predominantly planar circuits include **microstrip, stripline, coplanar waveguide**, PCB (single and multiple layers)



How Sonnet works

- Use waveguide modes of conductive box to expand the field in terms of current.
- Invert the matrix to obtain currents everywhere with respect to boundary conditions on the field.



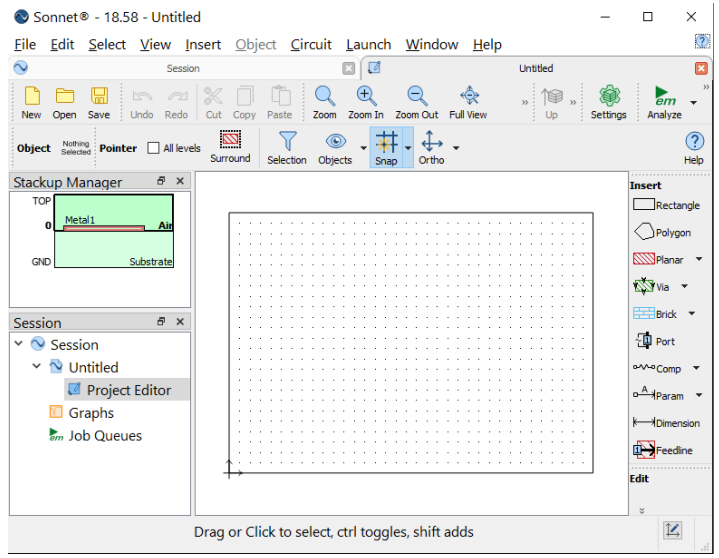
Circuit to simulate in a box [1]

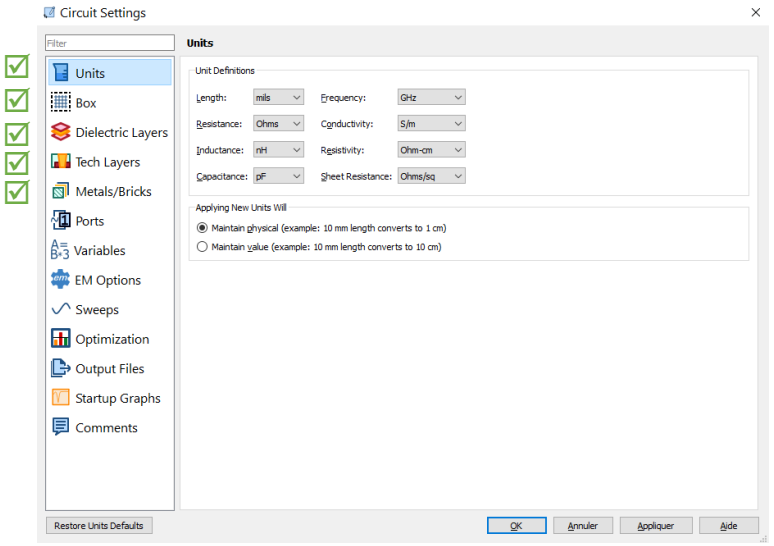
[1] J. C. Rautio and R. F. Harrington, "An Electromagnetic Time-Harmonic Analysis of Shielded Microstrip Circuits", IEEE Trans. Microwave Theory Tech., Vol. MTT-35, Aug. 1987, pp. 726-730.

Intermezzo : example of transmission line

1. Set Up the Project (Open Sonnet Software and create a new project)
    a. create a blank project
    b. save the project with an appropriate name (Microstrip.son).
2. Configure the Substrate

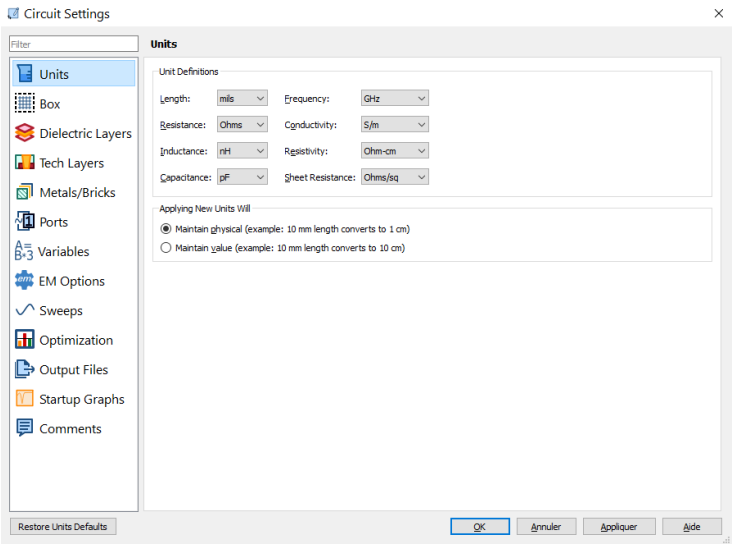
    a. Define the layers in your simulation:
        - Metallic Layer: Specify the properties of the metal (e.g., lossless, resistive, or finite conductivity).
        - Dielectric Layers: Define the substrate (e.g., silicon with thickness $h = 280\ \mu m$) and other dielectric layers like vacuum.
    b. Specify the thickness of each dielectric layer and its relative permittivity, metal properties (we start with lossless)





Intermezzo : example of transmission line

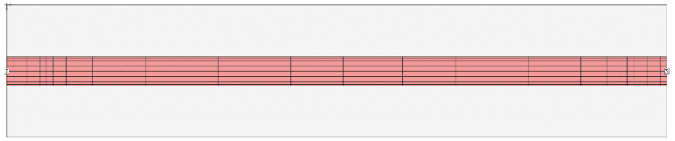
3. Use the Draw tools to create the geometry of the structure
   a. define the stripline (e.g., width $w \cong 225\ \mu m$ and length $L \cong 1\ mm$. `Insert ⇨ Metal ⇨ Rectangle`
   b. place the stripline at the appropriate location.
   c. ensure boundary conditions (e.g., box size, number of cells, symmetry....



5. Define Ports (add ports for excitation and measurement)
   ports can be edge ports or via ports depending on the design (see ref. plane).

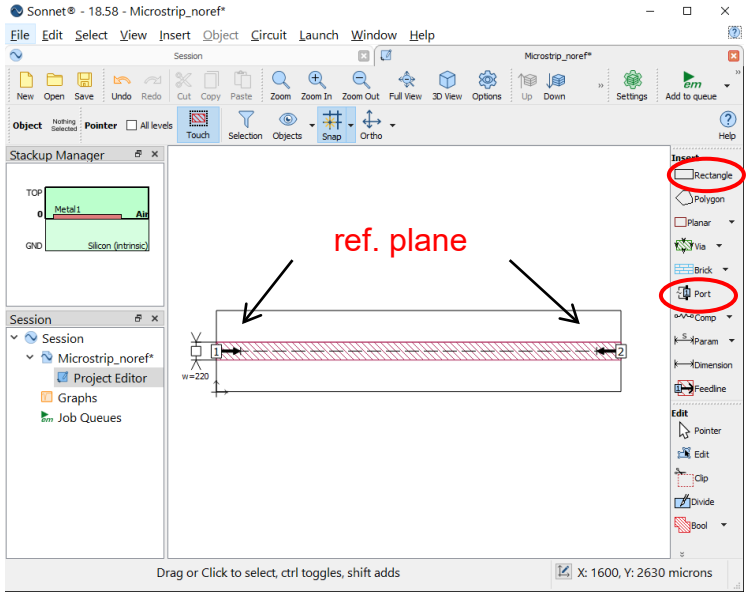
6. Variables `Insert ⇨ Dimension parameter ⇨ Symmetric`

7. Set simulation parameters (EM Options): compute current, estimate memory (specify accuracy levels), view subsection (mesh)…



8. Sweeps: choose the frequency sweep type (linear, adaptative, number of points), sweep parameters

9. Run the Simulation



Scikit-rf library

1. Importing scikit-rf : `ntwk = rf.Network('example.s2p')`
2. Working with Network Files
   a) Load a network file (Touchstone .s2p, .s1p, etc.): ntwk = `rf.Network('example.s2p')`
   b) Save a network file: `ntwk.write_touchstone('output.s2p')`
   c) Create a network from data:

```
freqs = rf.Frequency(1, 10, 101, unit='ghz')
s_params = ...  # S-parameters array, shape (freqs, ports, ports)
ntwk = rf.Network(frequency=freqs, s=s_params)
```

   d) Access network parameters:

```
ntwk.s      # S-parameters (complex matrix)
ntwk.f      # Frequencies
ntwk.z0     # Port impedance
```

3. Frequency Management
   a) Create a frequency object: `freq = rf.Frequency(1, 10, 201, unit='ghz')  # 1-10 GHz, 201 points`
   b) Resample or slice frequencies:

```
ntwk['2-5ghz']      # Slices network from 2 GHz to 5 GHz
ntwk.interpolate(freq)   # Interpolates to new frequency points
```

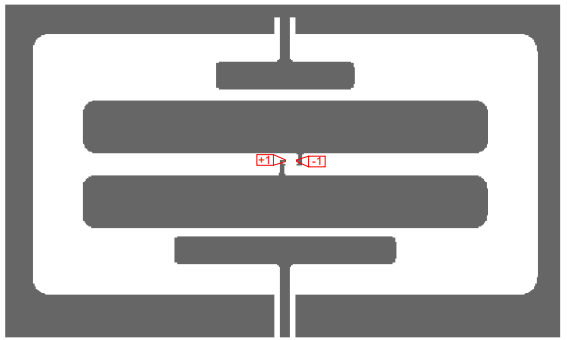
4. Network Properties

```
ntwk.s11   # Reflection coefficient at port 1
ntwk.s21   # Transmission coefficient from port 1 to port 2
```

Simulation of the Transmon Qubit

- We are looking for the value of the inductances L_g and L_e which simulate the qubit in the ground and excited states respectively.
- Charging energy:
  - Sonnet simulation enable to calculate de admittance of the capacitor in a given range of frequency by generating a snp file: `Qubit_Capa_2ports.s2p`.
  - Use the admittance to deduce the capacitance C and the charging energy E_c of the transmon.
  - Knowing the spectrum $E_n \simeq -E_\mathrm{J} + \sqrt{8E_\mathrm{J}E_\mathrm{c}}\left(n+\frac{1}{2}\right) - \frac{E_\mathrm{c}}{12}\left(6n^2+6n+3\right)$ and the frequency f_q of the qubit, we deduce its Josephson energy:

$$E_\mathrm{J} = \frac{(E_\mathrm{c}+hf_\mathrm{q})^2}{8E_\mathrm{c}}$$



  - Finally, the inductances are given by:

$$L_\mathrm{g} = \frac{L_\mathrm{J}}{\langle 0|\cos\phi|0\rangle} \simeq L_\mathrm{J}\left(1+\sqrt{\frac{E_\mathrm{c}}{2E_\mathrm{J}}}\right) \quad \text{and} \quad L_\mathrm{e} = \frac{L_\mathrm{J}}{\langle 1|\cos\phi|1\rangle} \simeq L_\mathrm{J}\left(1+3\sqrt{\frac{E_\mathrm{c}}{2E_\mathrm{J}}}\right)$$

  - Look at the resonance frequency of the LC resonators

Exercise: notebook Transmon

- Load 'Qubit_junction_capa.s1p' using scikit-rf ($0.1-10\ \mathrm{GHz}$, balanced ports ± 1)
- Extract the transmon capacitance
- Deduce the charging energy E_c
- Compute Josephson energy E_J for $f_q = 3.8\ \mathrm{GHz}$
- Determine inductances L_g and L_e ($|g\rangle$ and $|e\rangle$ states)
- Identify resonance frequencies from L_g and L_e simulations
- Plot S_{11} reflection at port 2
- Fit phase response: $\mathrm{Arg}\, S_{11} = \theta_0 + 2 \arctan\left[2Q_l\left(\frac{f_r - f}{f_r}\right)\right]$
- Extract coupling rate $\kappa_c = \frac{2\pi f_r}{Q_l}$

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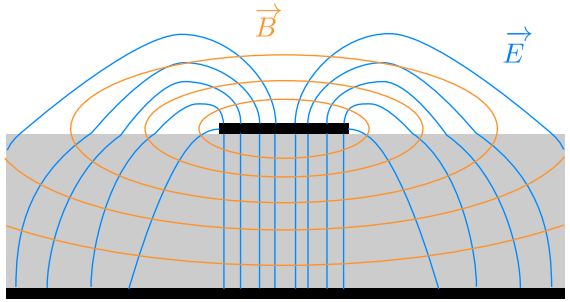
Propagation along a line

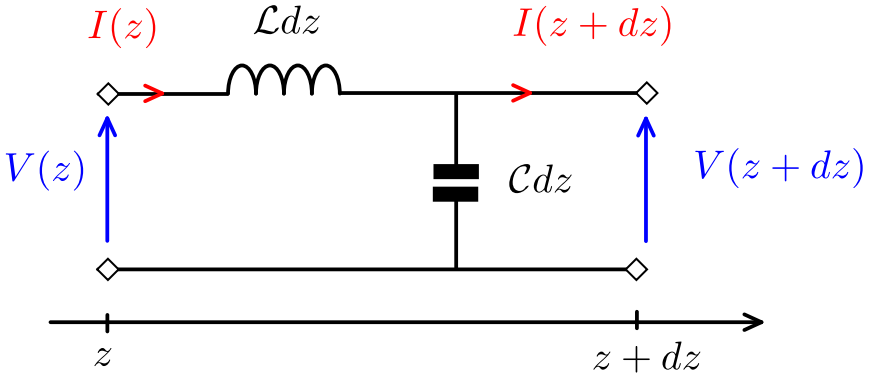
1. Two coupled variables that depend on time and space end up with a propagation (wave) equation because:
   - The coupling gives 2 first-order equations : the time variation of voltage $V(t,z)$ depends on the spatial gradient of current $I(z,t)$ and conversely: $\frac{\partial V}{\partial z} = -\mathcal{L}\frac{\partial I}{\partial t}$ and $\frac{\partial I}{\partial z} = -\mathcal{C}\frac{\partial V}{\partial t}$

   - By differentiating one of the equations and then using the second one to substitute for the other variable, you eliminate the “intermediate” variable and obtain a second-order equation in both time and space for a single variable — which is a propagation equation:

$$\frac{\partial^2 V}{\partial t^2} = \frac{1}{\mathcal{L}\mathcal{C}}\frac{\partial^2 V}{\partial z^2}$$



2. ID of a transmission line:
   - Characteristic impedance: $Z_0 = \pm\frac{V}{I} = \sqrt{\frac{\mathcal{L}}{\mathcal{C}}}$
   - Velocity: $v = \frac{1}{\sqrt{\mathcal{L}\mathcal{C}}}$
   - Inductance is not only electromagnetic, $\mathcal{L} = \mathcal{L}_{\mathrm{em}} + \mathcal{L}_{\mathrm{k}}$ it can also be kinetic (superconductors)



Exercise: notebook Microstrip.ipynb

- Plot the scattering parameters S_{11} and S_{21} for all transmission lines (the set of simulated frequencies can be extracted from the scikit-rf Network structure),
- From $\mathrm{Arg}(S_{11})$, infer the signal propagation velocity v along the microstrip,
- From the ABCD (transmission) matrix, extract the characteristic impedance Z_c of the transmission line.
- $\begin{pmatrix} V_1 \\ I_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} V_2 \\ I_2 \end{pmatrix}$ and for a transmission line $\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} \cos kL & iZ_c \sin kL \\ \frac{i}{Z_c} \sin kL & \cos kL \end{pmatrix}$ then $Z_c = \sqrt{\frac{B}{C}}$,
- Using v and Zc, derive the per-unit-length capacitance and inductance of the line (use the values corresponding to a kinetic-inductance sheet of $0.2\ pH/sq$),
- Recover the kinetic inductance as well as the electromagnetic (magnetic) inductance.

Quarter/half wave resonators

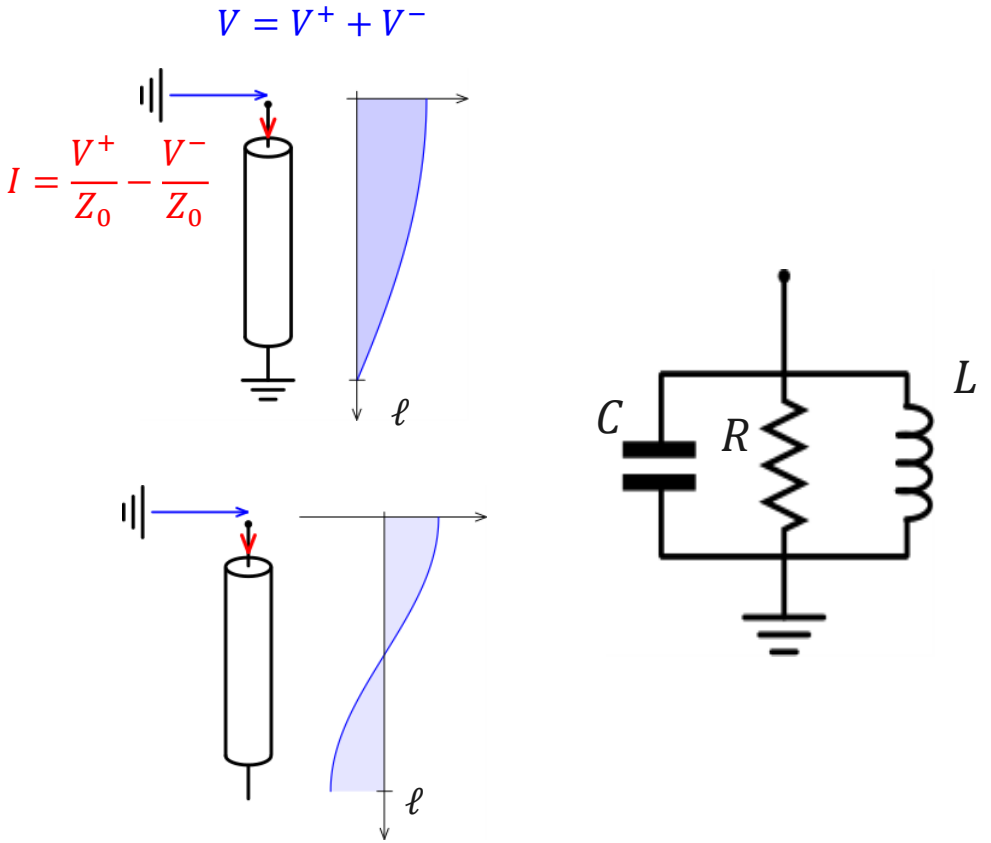
Stub filters: quarter-wave short-circuited or half-wave open-circuited transmission line stubs (~parallel resonant circuits)

a. Short-Circuited $\lambda/4$ Line

$$Z_{\text{in}} = Z_0 \,\text{th}\, k\,\ell \cong \frac{Z_0}{\alpha\ell + \text{j}\pi\frac{\Delta\omega}{2\omega_0}}$$

$\ell = \frac{\lambda}{4}$ at ω_0, $\Delta\omega = \omega - \omega_0$ and $k\ell = \frac{\pi}{2} + \pi\frac{\Delta\omega}{2\omega_0}$

$$Z_{\text{in}} = \frac{1}{\frac{1}{R} + j\frac{\pi}{2Z_0}\frac{\Delta\omega}{\omega_0}} \text{ with } R = \frac{Z_0}{\alpha\ell}, C = \frac{\pi}{4\omega_0 Z_0}, L = \frac{1}{\omega_0^2 C} \text{ and } Q_0 = \frac{\beta}{2\alpha}$$



b. Open-Circuited $\lambda/2$ Line

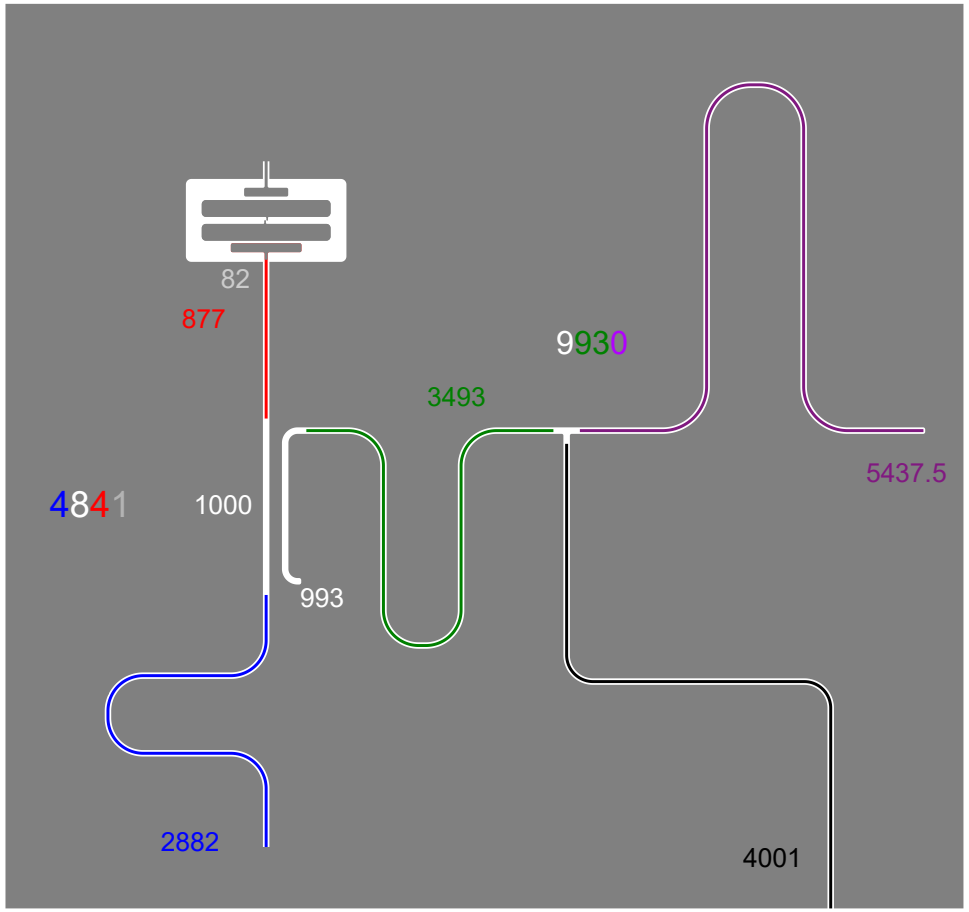
$$Z_{\text{in}} = Z_0 \coth k\,\ell k \cong \frac{Z_0}{\alpha\ell + \text{j}\pi\dfrac{\Delta\omega}{\omega_0}}$$

$\ell = \frac{\lambda}{2}$ at ω_0, $\Delta\omega = \omega - \omega_0$ and $\beta\ell = \pi + \pi\frac{\Delta\omega}{\omega_0}$

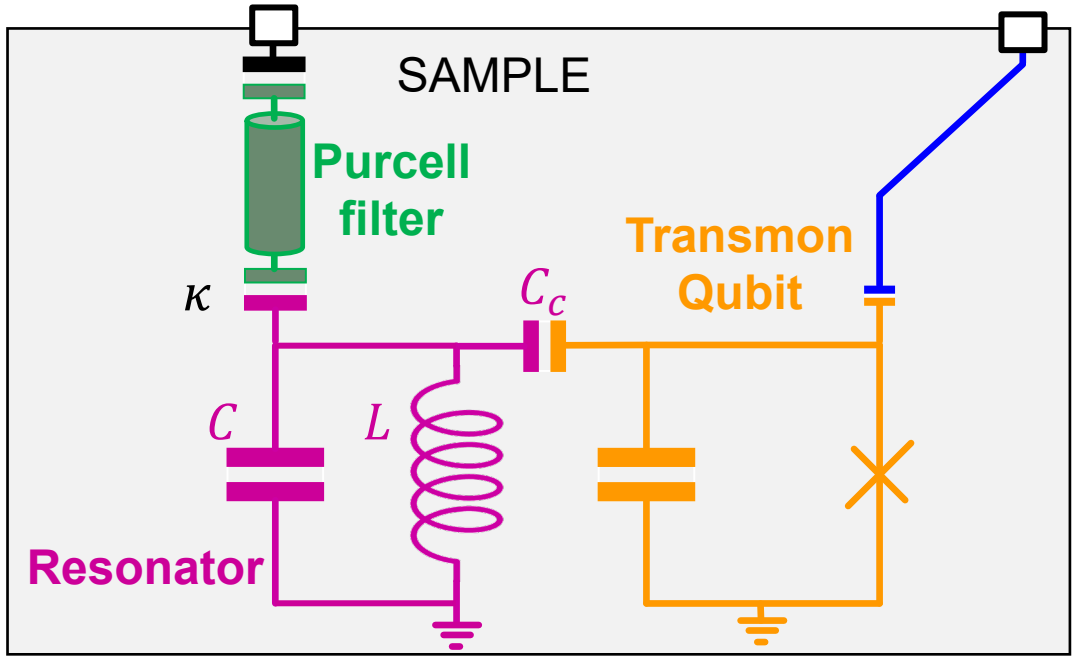
$$Z_{\text{in}} = \frac{1}{\frac{1}{R} + 2jC\Delta\omega} \text{ with } R = \frac{Z_0}{\alpha\ell}, C = \frac{\pi}{2\omega_0 Z_0}, L = \frac{1}{\omega_0^2 C} \text{ and } Q_0 = \frac{\beta}{2\alpha}$$

c. Quarter-wave impedance transformer: $\frac{Z_{\text{in}}}{Z_0} = \frac{Z_0}{Z_{\text{L}}}$

On the qubit board



- Readout resonator: $f_r = \frac{v}{4\ell_{\text{res}}} = \frac{1.26\times10^8}{4\times0{,}004841} \cong 6.51\ GHz,$
- Purcell filter: $f_p = \frac{v}{2\ell_{\text{res}}} = \frac{1.26\times10^8}{2\times0{,}010075} \cong 6.25\ GHz$



Sonnet scikit-rf – example of transmission line

1. Use the Jupyter server: https://jupyter.lps.u-psud.fr/
2. Importing scikit-rf : `ntwk = rf.Network('example.s2p')`
3. Working with Network Files
   a) Load a network file (Touchstone .s2p, .s1p, etc.): ntwk = `rf.Network('example.s2p')`
   b) Save a network file: `ntwk.write_touchstone('output.s2p')`
   c) Create a network from data:

```
freqs = rf.Frequency(1, 10, 101, unit='ghz')
s_params = ...  # S-parameters array, shape (freqs, ports, ports)
ntwk = rf.Network(frequency=freqs, s=s_params)
```

   d) Access network parameters:

```
ntwk.s      # S-parameters (complex matrix)
ntwk.f      # Frequencies
ntwk.z0     # Port impedance
```

4. Frequency Management
   a) Create a frequency object: `freq = rf.Frequency(1, 10, 201, unit='ghz')` *# 1-10 GHz, 201 points*
   b) Resample or slice frequencies:

```
ntwk['2-5ghz']      # Slices network from 2 GHz to 5 GHz
ntwk.interpolate(freq)   # Interpolates to new frequency points
```

4. Network Properties

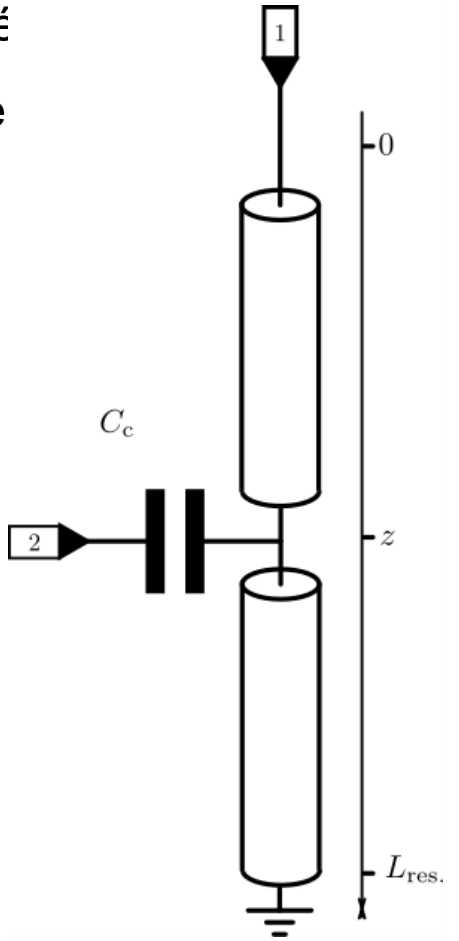
```
ntwk.s11    # Reflection coefficient at port 1
ntwk.s21    # Transmission coefficient from port 1 to port 2
```

Exercise: notebook Purcell

- Crée des lignes de transmission skrf définies par une longueur $L_{\text{Res.}} - z$ et z, une impé caractéristique $Z_0 = 54.93\ \Omega$ et une vitesse de propagation $v = 1.26 \times 10^{8}\text{^}8\ \text{m} \cdot \text{s}^{-1}$ e aidant de l'exemple ci-dessous :

```
s_line = np.zeros((Nf, 2, 2), dtype=complex)
s_line[:, 0, 1] = np.exp(-1j * 2 * np.pi * f / c * L)
s_line[:, 1, 0] = np.exp(-1j * 2 * np.pi * f / c * L)
CPW = rf.Network(frequency=freq, s=s_line, z0=50, name='CPW_a_b')
```

- Créer 2 ports, un short et un élément de circuit discret simulant la capacité C_c.
- Mesurer la fréquence de résonance f_0 sachant que $\text{Im}\ \text{Y}(f_0) = 0$,
- Créer des connexions simulant le circuit ci-dessus pour différentes valeurs de ℓ.

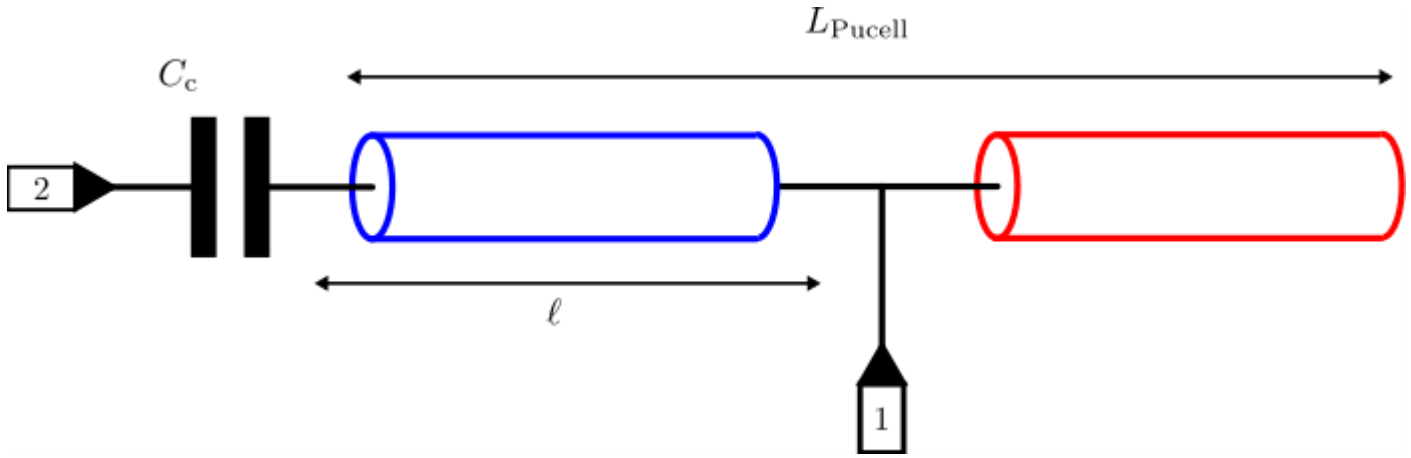


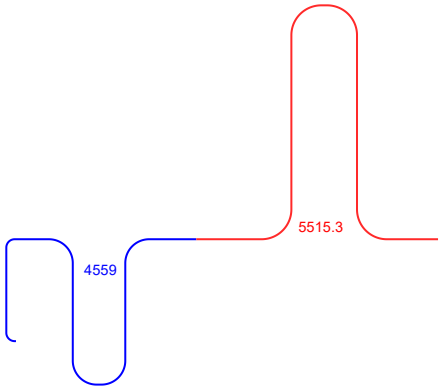
Exercise: notebook Purcell

- Crée des lignes de transmission skrf définies par une longueur $L_{\text{Purcell}} - \ell$ et ℓ, une impédance caractéristique $Z_0 = 54.93\ \Omega$ et une vitesse de propagation $v = 1.26 \times 10^8$^8 $\text{m} \cdot \text{s}^{-1}$ en vous aidant de l'exemple ci-dessous :

```
s_line = np.zeros((Nf, 2, 2), dtype=complex)
s_line[:, 0, 1] = np.exp(-1j * 2 * np.pi * f / c * L)
s_line[:, 1, 0] = np.exp(-1j * 2 * np.pi * f / c * L)
CPW = rf.Network(frequency=freq, s=s_line, z0=50, name='CPW_a_b')
```

- Créer 2 ports, un open et un élément de circuit discret simulant la capacité C_c.
- Créer des connexions simulant le circuit ci-dessus pour différentes valeurs de ℓ.



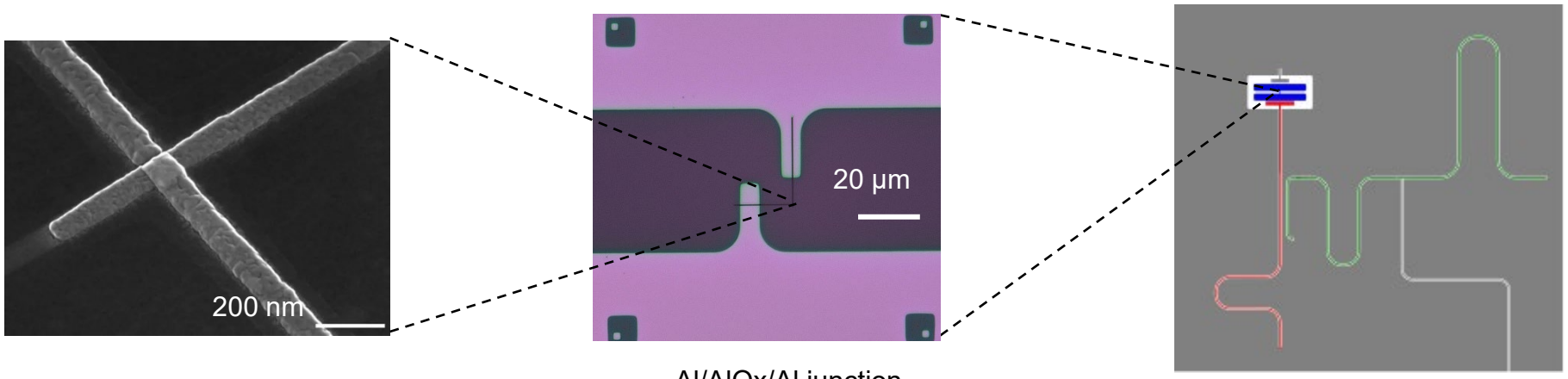


Dispersive Measurement of a Qubit

- Dispersive measurement is a quantum non-demolition (QND) technique used to determine the state of a qubit by observing its effect on a coupled resonator or cavity. This method minimizes back-action on the qubit, allowing repeated measurements without collapsing the qubit state.

- The system typically consists of a superconducting qubit (e.g., Aluminum transmon) capacitively coupled to a microwave resonator. In the dispersive regime, the qubit and resonator are detuned, so they do not exchange energy directly. Nevertheless, coupling to the resonator opens a radiative loss channel (Purcell effect), and a Purcell filter is introduced to suppress this decay without degrading readout performance (the role of the Purcell filter is to shape the impedance seen by the qubit). Instead, the qubit state shifts the resonator frequency by a small amount, known as the dispersive shift χ.

- The qubit is realized on a sapphire substrate ($\epsilon_{r,z} = 11.5$), resonator is in tantalum ($L_k \approx 0.2\ \mathrm{pH/sq}, T_c \approx 4.4\ \mathrm{K}$) and Josephson junction is aluminum made ($R_n \cong 10\ k\Omega$).



Al/AlOx/Al junction

Dispersive Measurement of a Qubit

– The qubit's state (e.g., $|0\rangle$ or $|1\rangle$) changes the effective resonance frequency of the resonator by $\pm\chi$. This shift is detected by sending a microwave probe tone through the resonator and measuring the phase or amplitude of the transmitted/reflected signal. The dispersive shift is given by:

$$\chi = \frac{g^2}{\Delta}$$

g is the coupling strength, κ_c the coupling rate and $\Delta = |\omega_r - \omega_q|$ is the detuning between the qubit and resonator.

– In the dispersive regime ($\Delta \gg g$) the Jaynes-Cummings Hamiltonian :

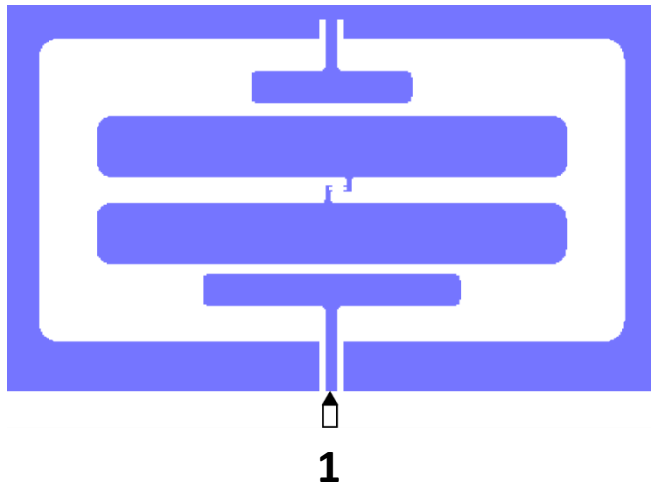
$$H_{\mathrm{JC}} = \hbar\omega_{\mathrm{r}} a^\dagger a + \frac{\hbar\omega_{\mathrm{q}}}{2}\sigma_z + \hbar g\left(a^\dagger\sigma_- + a\sigma_+\right)$$ becomes a QND Hamiltonian $[H_{\mathrm{disp.}}, \sigma_z] = 0$

$$H_{\mathrm{disp.}} = \hbar\left(\omega_{\mathrm{r}} + \chi\sigma_z\right) a^\dagger a + \hbar\frac{\omega_{\mathrm{q}} + \chi}{2}\sigma_z$$

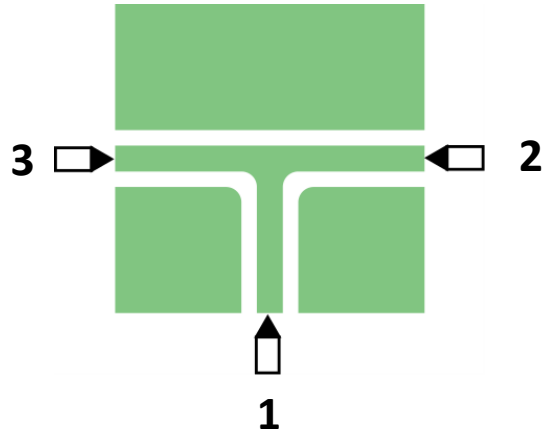
– readout frequency : $\omega_{\mathrm{r}} + \chi$ for $|1\rangle$ and $\omega_{\mathrm{r}} - \chi$ for $|0\rangle$

Simulated elements

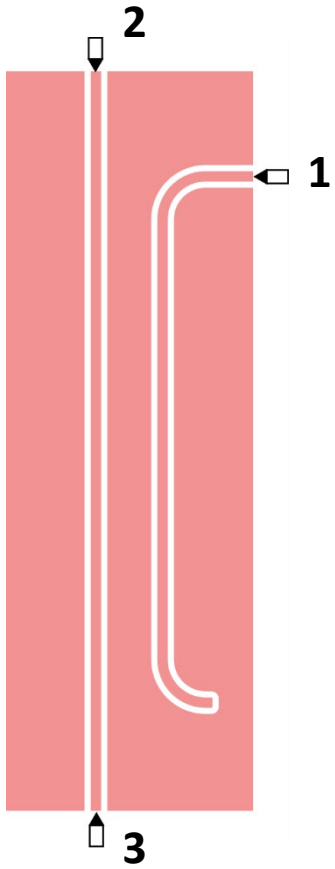
The qubit is realized on a sapphire substrate ($\epsilon_{r,z} = 11.5$), resonator is in tantalum ($L_k \approx 0.2\ \mathrm{pH/sq}, T_c \approx 4.4\ \mathrm{K}$) and Josephson junction is aluminum made ($R_n \cong 10\ k\Omega$).
Sonnet simulation (Cell size: 2.5μ X 2.5 μm and a bandwidth (10000 points) [3 GHz, 7 GHz]) gives .snp files



Qubit_junction_g.s1p
Qubit_junction_e.s1p



Qubit_T_connection.s3p

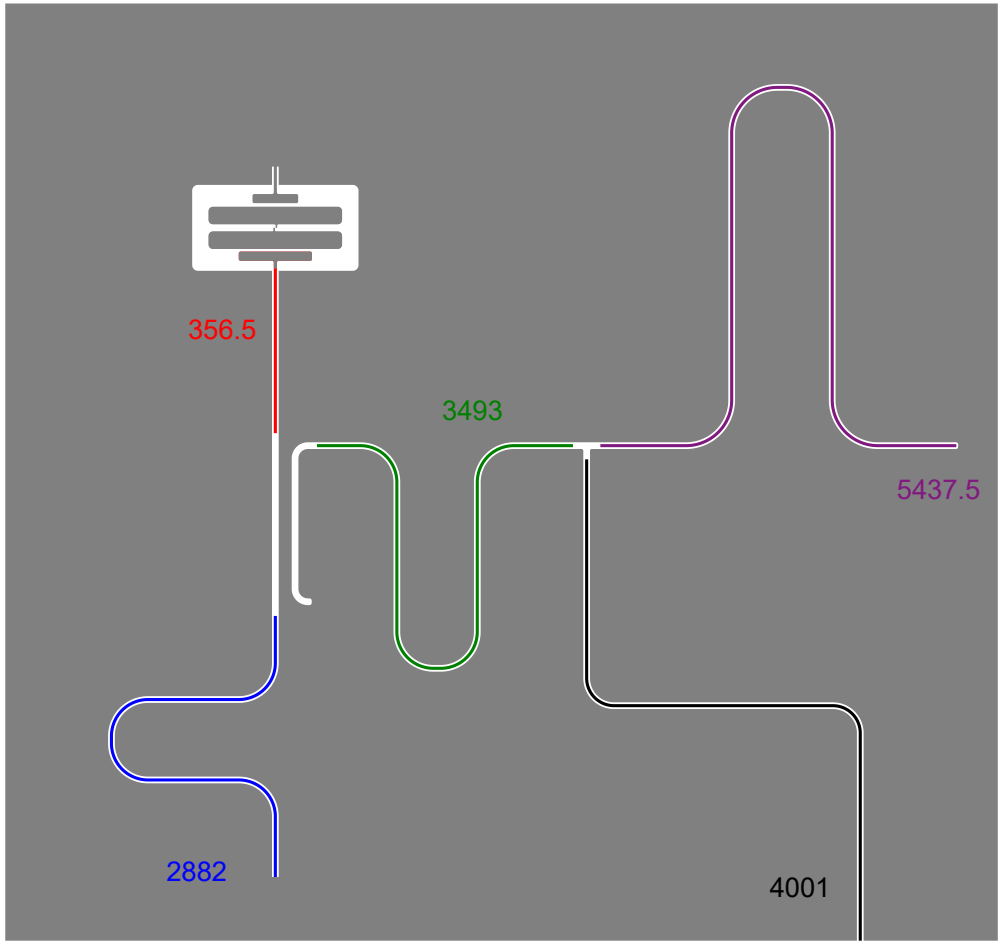


Qubit_coupler.s3p

Length of different lines

Length of different CPW lines in tantal ($Z_c = 54.93\ \Omega$ and $v = 1{,}26 \times 10^8\ \mathrm{m.s^{-1}}$)

- $L_{\mathrm{port-T}} = 4001\ \mu m$
- $L_{\mathrm{T-open}} = 5437.3 \mu m$
- $L_{\mathrm{coupler-T}} = 3493 \mu m$
- $L_{\mathrm{coupler-Qubit}} = 356.5\ \mu m$
- $L_{\mathrm{coupler-short}} = 2882\ \mu m$
- $L_{\mathrm{resonator}} = 4841\ \mu\mathrm{m}$
- $L_{\mathrm{Purcell}} = 10075\ \mu\mathrm{m}$



Simulation of the CPW lines of the Qubit

Write a Python script using the following libraries:

```
import matplotlib.pyplot as plt
import numpy as np
import skrf as rf
```

- Create the 5 scikit CPW_coupler Network like:

```
s_coupler_Qubit = np.zeros((Nf, 2, 2), dtype=complex)
s_coupler_Qubit[:, 0, 1] = np.exp(-1j * 2 * np.pi * f / c * L_coupler_Qubit)
s_coupler_Qubit[:, 1, 0] = np.exp(-1j * 2 * np.pi * f / c * L_coupler_Qubit)
CPW_coupler_Qubit = Network(frequency=freq, s=s_coupler_Qubit, z0=50, name='CPW coupler_Qubit')
```

```
print(CPW_coupler_Qubit)
```

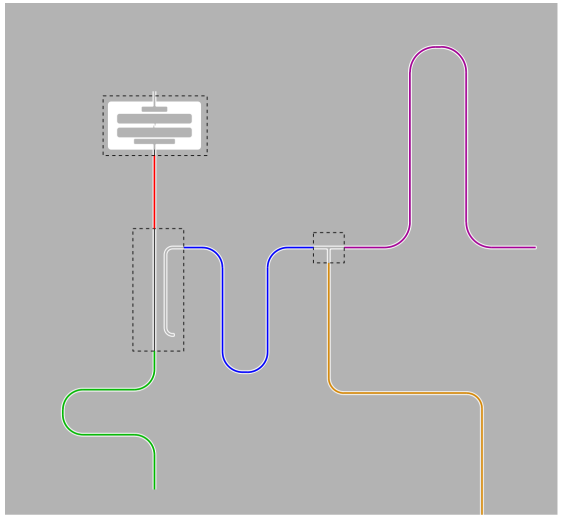
```
2-Port Network: 'CPW coupler_Qubit',  4.0-7.0 GHz, 60001 pts, z0=[50.+0.j 50.+0.j]
```

- Create the circuit with a DUT simulating the ground and excited states of the qubit:

```
port1 = rf.Circuit.Port(frequency=freq, name='port1', z0=50)
port2 = rf.Circuit.Port(frequency=freq, name='port2', z0=50)
ground0 = rf.Circuit.Ground(freq, name='GND0')
line = rf.media.DefinedGammaZ0(frequency=freq, z0=50)
Open1 =line.resistor(1e9, name='open1') # end of the Purcell filter
```

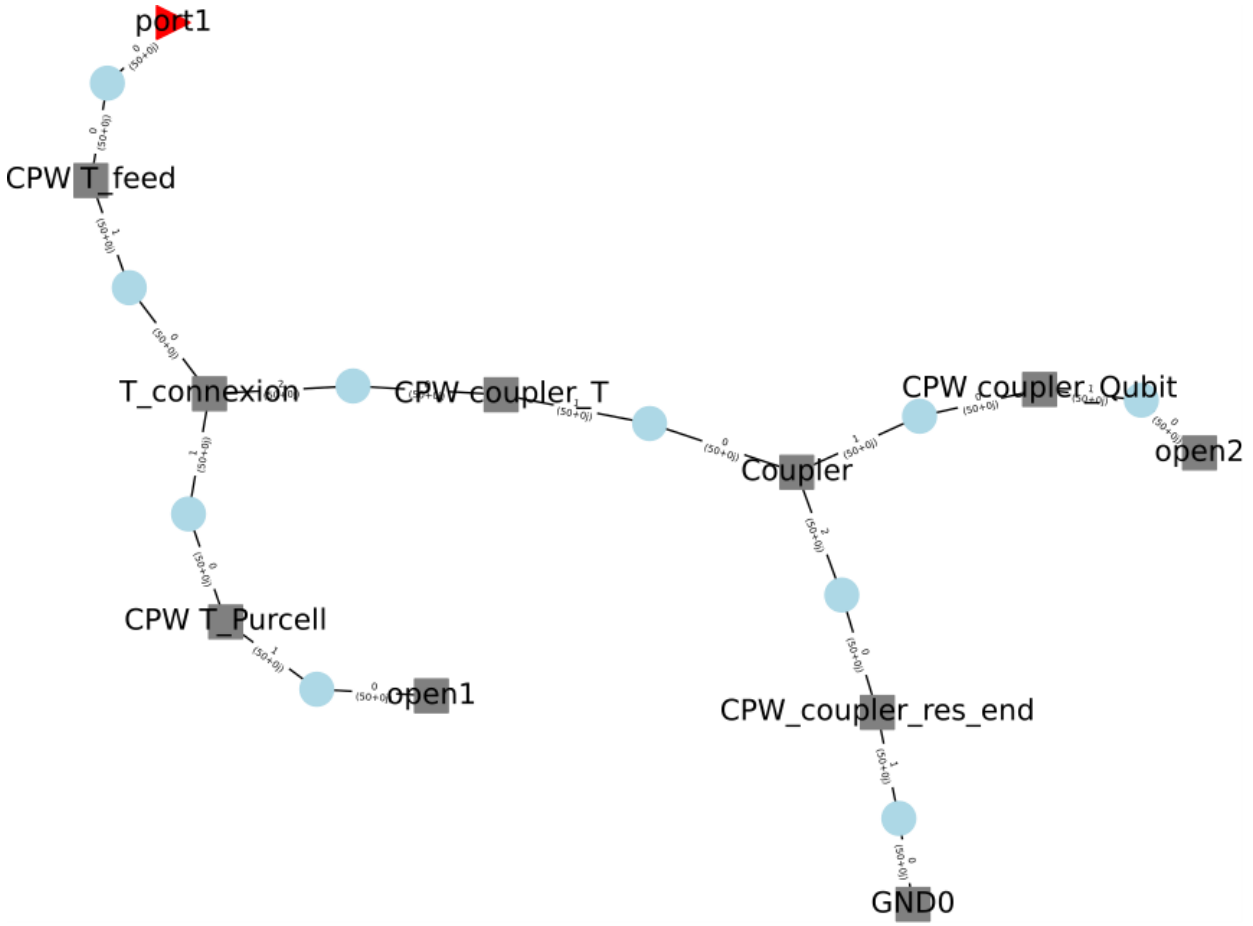
```
DUT=[]
DUT.append(Qubit_g) # ground state (Junction_param2.n1p)
DUT.append(Qubit_e) # excited state (Junction_param3.n1p)
```

```
connexion=[]
for k in range(2):
    connexion.append([[(port1, 0), (CPW_T_feed, 0)],
                      [(CPW_T_feed, 1), (T_conn,0)],
                      [(T_conn,1),(CPW_T_Purcell,0)],
                      [(CPW_T_Purcell, 1),(Open1,0)],
                      [(T_conn,2),(CPW_coupler_T,0)],
                      [(CPW_coupler_T,1),(coupler,0)],
                      [(coupler,1),(CPW_coupler_Qubit,0)],
                      [(CPW_coupler_Qubit,1),(DUT[k],0)],
                      [(coupler,2),(CPW_coupler_res_end,0)],
                      [(CPW_coupler_res_end,1),(ground0,0)]
                     ])
resulting_Circuit=[]
```



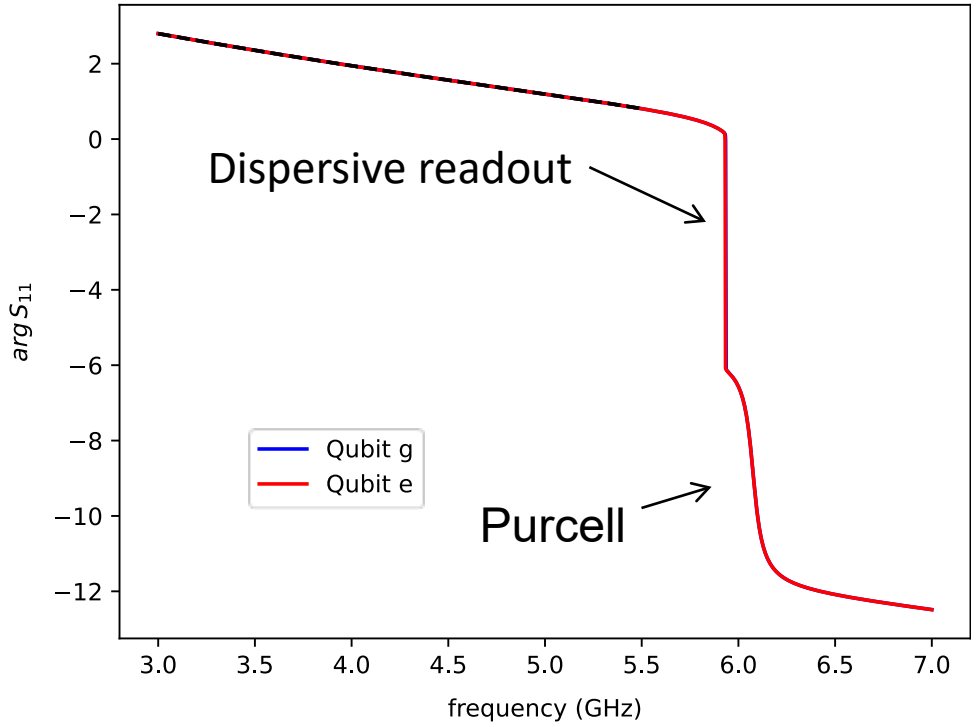
Transmon Qubit

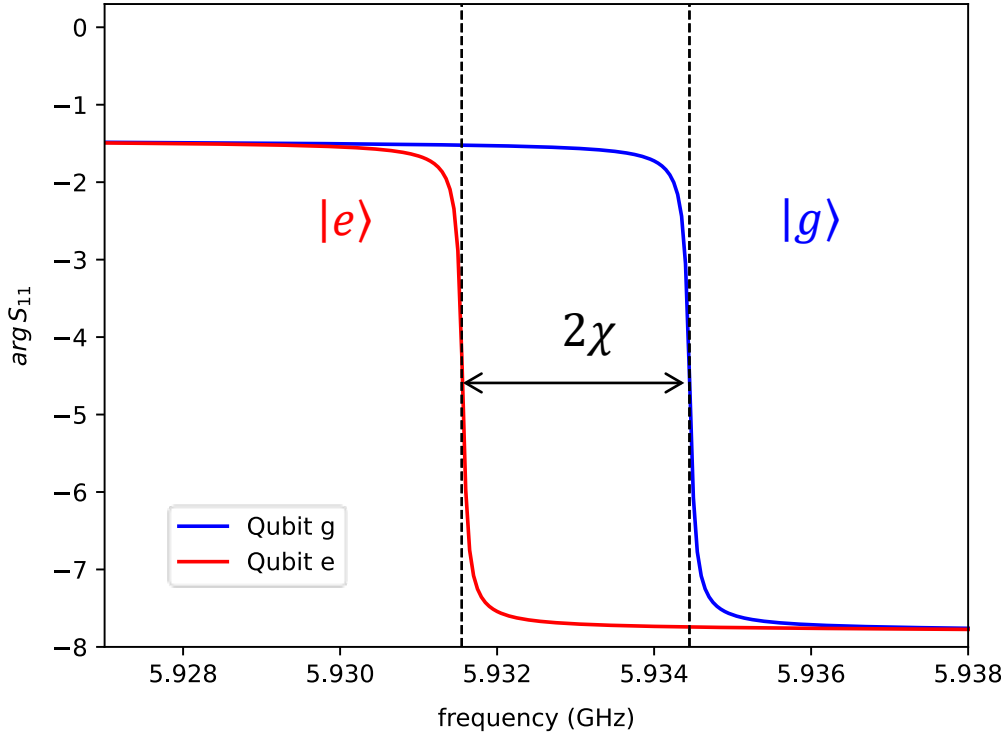
`scikit-rf` circuit



Transmon Qubit with Purcell filter

`scikit-rf` circuit: open or qubit ended





Dispersive readout ($\Delta = (\omega_r - \omega_q) \gg g, \kappa$): $\omega_r = 2\pi \times 6.50$ GHz and $\omega_q = 2\pi \times 3.8$ GHz

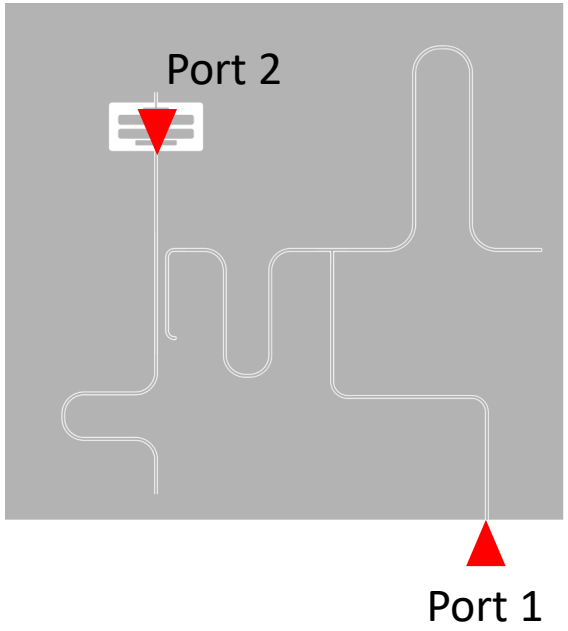
$$H_{J.C.} = \hbar\omega_r\left(a^\dagger a + \frac{1}{2}\right) + \frac{\hbar\omega_q}{2}\sigma_z + g\left(\sigma_+ a + \sigma_- a^\dagger\right) \Rightarrow H_{disp.} = (\hbar\omega_r + \chi\sigma_z)\left(a^\dagger a + \frac{1}{2}\right) + \frac{\hbar\tilde{\omega}_q}{2}$$

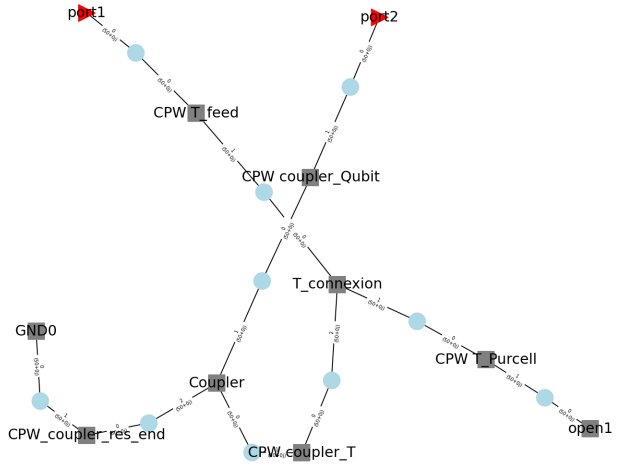
$$\tilde{\omega}_q = \omega_q + \frac{g^2}{\Delta} \qquad \chi = \frac{g^2}{\Delta}$$

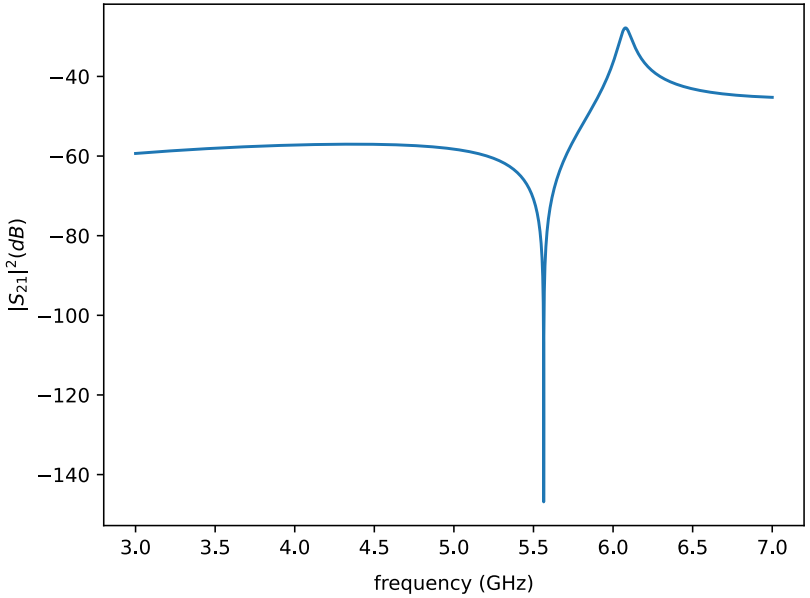
We find $\chi \approx 1.42\ MHz$ and with $\Delta = \omega_r - \omega_q \approx 2\pi \times 2.70$ GHz then $g \approx 63$ MHz to be compared to $\kappa_c = 52.9$ MHz .

Effect of the Purcell filter

`scikit-rf` circuit: we consider a second port before the coupling to the cavity





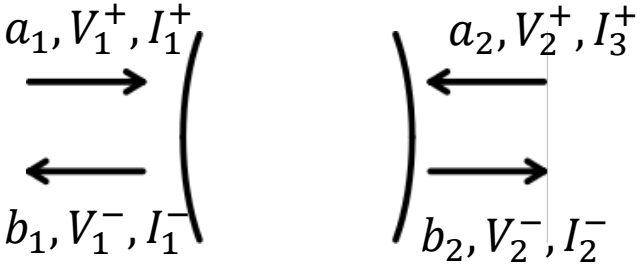


$f_{\mathrm{eg}} \approx 4\ \mathrm{GHz}$

2-port microwave network S,T,(ABCD) matrices

- Scattering matrix $[S]$: the measured quantities due to the nature of wave propagation in transmission lines

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$



- Transmission matrix $[T]$: simplifies cascading analysis of multi-stage networks

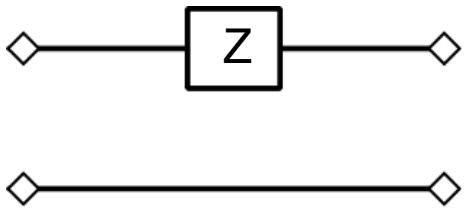
$$\begin{pmatrix} b_1 \\ a_1 \end{pmatrix} = \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix} \begin{pmatrix} a_2 \\ b_2 \end{pmatrix}$$

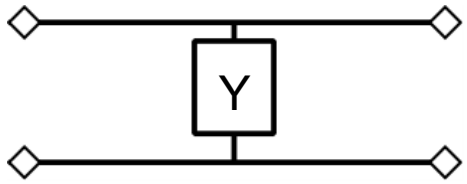
$$a_i = \frac{V_i^+}{\sqrt{Z_0}}, \; b_i = \frac{V_i^-}{\sqrt{Z_0}}$$

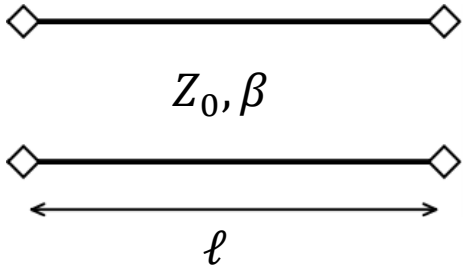
- Transmission (ABCD) matrix

$$\begin{pmatrix} V_1 \\ I_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} V_2 \\ I_2 \end{pmatrix}$$

Exercise: Calculate the transmission matrix for the following Two-Port Circuits

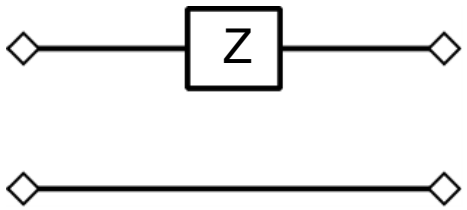






2-port microwave network S,T,(ABCD) matrices

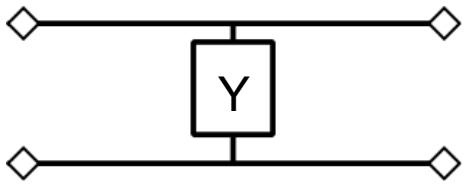
- Solution



$$V_1^+ + V_1^- = V_2^+ + V_2^- + Z(I_2^+ + I_2^-)$$
$$I_1^+ + I_1^- = I_2^+ + I_2^-$$

$$[T(Z)] = \begin{bmatrix} 1 - \frac{Z}{2Z_0} & \frac{Z}{2Z_0} \\ -\frac{Z}{2Z_0} & 1 + \frac{Z}{2Z_0} \end{bmatrix}$$
$$T(Z_1 + Z_2) = T(Z_1) \times T(Z_2)$$

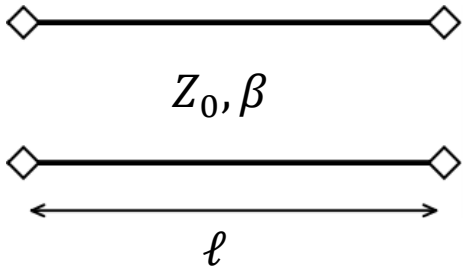
$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$$



$$V_1^+ + V_1^- = V_2^+ + V_2^-$$
$$I_1^+ + I_1^- = I_2^+ + I_2^- + Y(V_2^+ + V_2^-)$$

$$[T(Y)] = \begin{bmatrix} 1 - \frac{Z_0 Y}{2} & \frac{Z_0 Y}{2} \\ -\frac{Z_0 Y}{2} & 1 + \frac{Z_0 Y}{2} \end{bmatrix}$$
$$T(Y_1 + Y_2) = T(Y_1) \times T(Y_2)$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix}$$



$$V_1^+ + V_1^- = e^{-j\beta} V_2^+ + e^{j\beta} V_2^-$$
$$I_1^+ + I_1^- = e^{-j\beta} I_2^+ + e^{j\beta} I_2^-$$

$$[T(Y)] = \begin{bmatrix} e^{-j\beta} & 0 \\ 0 & e^{j\beta} \end{bmatrix}$$

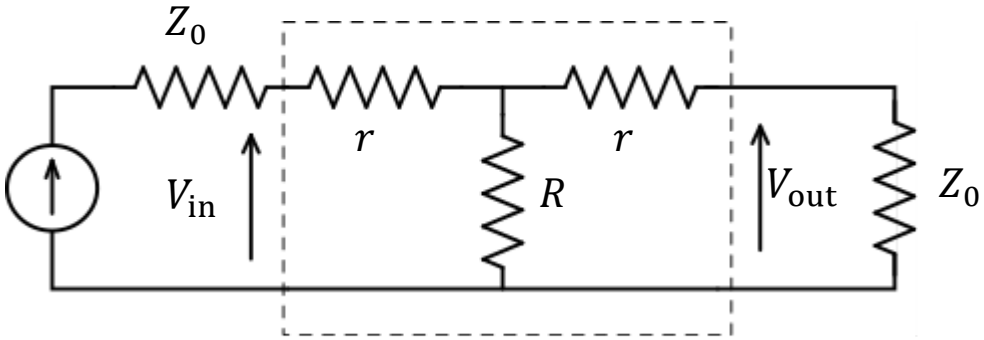
$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \cos\beta\ell & jZ_0 \sin\beta\ell \\ \frac{j}{Z_0}\sin\beta\ell & \cos\beta\ell \end{bmatrix}$$

Conversion between Two-Port Network Parameters

	S	Z	Y	$ABCD$
S_{11}	S_{11}	$\frac{(Z_{11}-Z_0)(Z_{22}+Z_0)-Z_{12}Z_{21}}{\Delta Z}$	$\frac{(Y_0-Y_{11})(Y_0+Y_{22})+Y_{12}Y_{21}}{\Delta Y}$	$\frac{A+B/Z_0-CZ_0-D}{A+B/Z_0+CZ_0+D}$
S_{12}	S_{12}	$\frac{2Z_{12}Z_0}{\Delta Z}$	$\frac{-2Y_{12}Y_0}{\Delta Y}$	$\frac{2(AD-BC)}{A+B/Z_0+CZ_0+D}$
S_{21}	S_{21}	$\frac{2Z_{21}Z_0}{\Delta Z}$	$\frac{-2Y_{21}Y_0}{\Delta Y}$	$\frac{2}{A+B/Z_0+CZ_0+D}$
S_{22}	S_{22}	$\frac{(Z_{11}+Z_0)(Z_{22}-Z_0)-Z_{12}Z_{21}}{\Delta Z}$	$\frac{(Y_0+Y_{11})(Y_0-Y_{22})+Y_{12}Y_{21}}{\Delta Y}$	$\frac{-A+B/Z_0-CZ_0+D}{A+B/Z_0+CZ_0+D}$
Z_{11}	$Z_0\frac{(1+S_{11})(1-S_{22})+S_{12}S_{21}}{(1-S_{11})(1-S_{22})-S_{12}S_{21}}$	Z_{11}	$\frac{Y_{22}}{\lvert Y\rvert}$	$\frac{A}{C}$
Z_{12}	$Z_0\frac{2S_{12}}{(1-S_{11})(1-S_{22})-S_{12}S_{21}}$	Z_{12}	$\frac{-Y_{12}}{\lvert Y\rvert}$	$\frac{AD-BC}{C}$
Z_{21}	$Z_0\frac{2S_{21}}{(1-S_{11})(1-S_{22})-S_{12}S_{21}}$	Z_{21}	$\frac{-Y_{21}}{\lvert Y\rvert}$	$\frac{1}{C}$
Z_{22}	$Z_0\frac{(1-S_{11})(1+S_{22})+S_{12}S_{21}}{(1-S_{11})(1-S_{22})-S_{12}S_{21}}$	Z_{22}	$\frac{Y_{11}}{\lvert Y\rvert}$	$\frac{D}{C}$
Y_{11}	$Y_0\frac{(1-S_{11})(1+S_{22})+S_{12}S_{21}}{(1+S_{11})(1+S_{22})-S_{12}S_{21}}$	$\frac{Z_{22}}{\lvert Z\rvert}$	Y_{11}	$\frac{D}{B}$
Y_{12}	$Y_0\frac{-2S_{12}}{(1+S_{11})(1+S_{22})-S_{12}S_{21}}$	$\frac{-Z_{12}}{\lvert Z\rvert}$	Y_{12}	$\frac{BC-AD}{B}$
Y_{21}	$Y_0\frac{-2S_{21}}{(1+S_{11})(1+S_{22})-S_{12}S_{21}}$	$\frac{-Z_{21}}{\lvert Z\rvert}$	Y_{21}	$\frac{-1}{B}$
Y_{22}	$Y_0\frac{(1+S_{11})(1-S_{22})+S_{12}S_{21}}{(1+S_{11})(1+S_{22})-S_{12}S_{21}}$	$\frac{Z_{11}}{\lvert Z\rvert}$	Y_{22}	$\frac{A}{B}$
A	$\frac{(1+S_{11})(1-S_{22})+S_{12}S_{21}}{2S_{21}}$	$\frac{Z_{11}}{Z_{21}}$	$\frac{-Y_{22}}{Y_{21}}$	A
B	$Z_0\frac{(1+S_{11})(1+S_{22})-S_{12}S_{21}}{2S_{21}}$	$\frac{\lvert Z\rvert}{Z_{21}}$	$\frac{-1}{Y_{21}}$	B
C	$\frac{1}{Z_0}\frac{(1-S_{11})(1-S_{22})-S_{12}S_{21}}{2S_{21}}$	$\frac{1}{Z_{21}}$	$\frac{-\lvert Y\rvert}{Y_{21}}$	C
D	$\frac{(1-S_{11})(1+S_{22})+S_{12}S_{21}}{2S_{21}}$	$\frac{Z_{22}}{Z_{21}}$	$\frac{-Y_{11}}{Y_{21}}$	D

$\lvert Z\rvert = Z_{11}Z_{22}-Z_{12}Z_{21}$; $\lvert Y\rvert = Y_{11}Y_{22}-Y_{12}Y_{21}$; $\Delta Y = (Y_{11}+Y_0)(Y_{22}+Y_0)-Y_{12}Y_{21}$; $\Delta Z = (Z_{11}+Z_0)(Z_{22}+Z_0)-Z_{12}Z_{21}$; $Y_0 = 1/Z_0$.

Example: T-attenuator



The attenuator is matched to Z_0 and is characterized by an attenuation factor g:

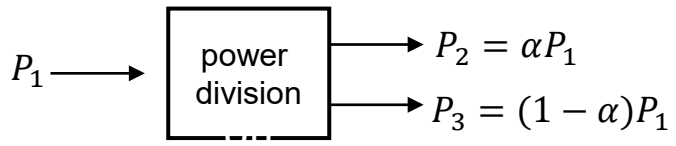
$$\begin{cases} Z_0 = \mathrm{r} + \dfrac{R(r+Z_0)}{R+r+Z_0} \\ g = \dfrac{Z_0}{r+Z_0}\dfrac{\dfrac{R(r+Z_0)}{R+r+Z_0}}{r+\dfrac{R(r+Z_0)}{R+r+Z_0}} \end{cases}$$

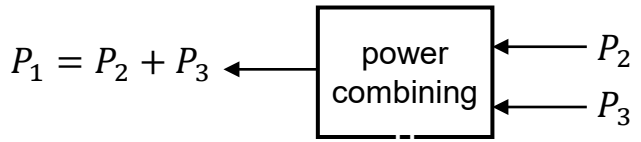
then

$$\begin{cases} r = \left(\dfrac{1-g}{1+g}\right)Z_0 \\ R = \left(\dfrac{2g}{1-g^2}\right)Z_0 \end{cases}$$

Refection measurement

1. Power Dividers and Directional Couplers:
2. Basic properties of dividers and couplers:
   a. Three-Port Networks (T-Junctions):

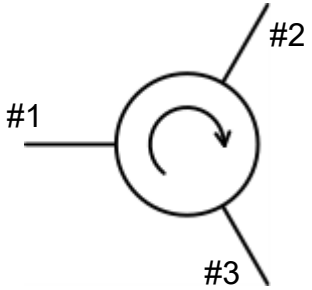




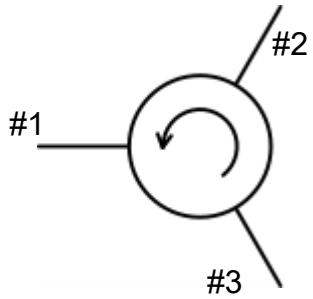
The simplest type of power divider is a *T-junction* of three ports described by $[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{33} & S_{32} & S_{33} \end{bmatrix}$

- if all ports are matched: $S_{ii} = 0, \forall i$, if the network is reciprocal: $S_{ij} = S_{ji}\ , \forall i, j$ and lossless: $S^{\dagger}S = I$: such a matrix does not exist.
- We have to relax the reciprocity constraint. We get the circulator

$$[S] = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

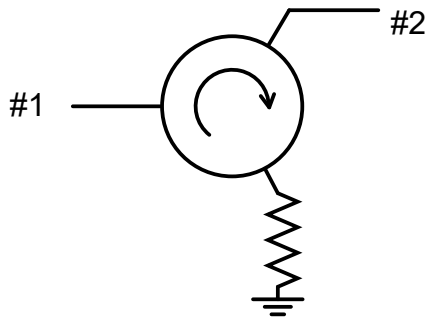


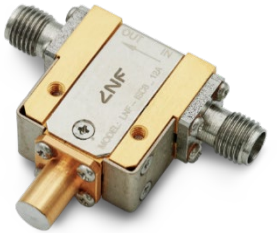
$$[S] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$



or insulator

$$[S] = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

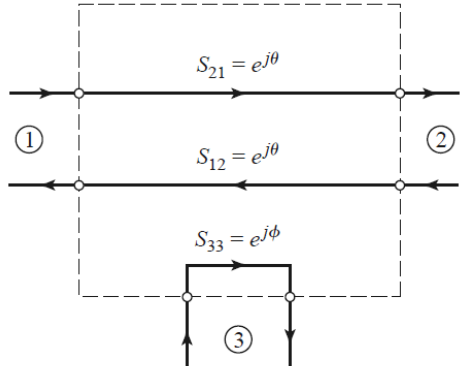




Refection measurement

- or relax the matching constraint on one port

$$[S] = \begin{bmatrix} 0 & e^{j\theta} & 0 \\ e^{j\theta} & 0 & 0 \\ 0 & 0 & e^{j\phi} \end{bmatrix}$$



- or relax the lossless constraint: example of the amplitude divider (-6 dB resistive divider)

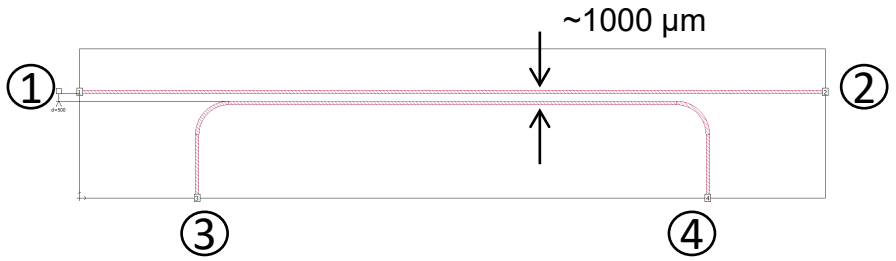
b. Four-Port Networks (Directional Couplers)

$[S] = \begin{bmatrix} 0 & S_{12} & S_{13} & S_{14} \\ S_{12} & 0 & S_{23} & S_{24} \\ S_{13} & S_{23} & 0 & S_{34} \\ S_{14} & S_{24} & S_{34} & 0 \end{bmatrix}$ for a matched, reciprocal and lossless coupler, we get $\begin{vmatrix} |S_{12}|^2 + |S_{13}|^2 = 1 \\ |S_{12}|^2 + |S_{24}|^2 = 1 \\ |S_{13}|^2 + |S_{34}|^2 = 1 \\ |S_{24}|^2 + |S_{34}|^2 = 1 \end{vmatrix}$ and by choosing a phase reference on ports 3 and 4, we note: $S_{12} = \alpha$ and $S_{13} = \beta e^{i\theta}$ and $S_{24} = \beta e^{i\varphi}$. From unitarity ${S_{12}}^* S_{13} + {S_{24}}^* S_{34} = 0$, we get $\theta + \varphi = \pi$

Symmetric: $\theta = \varphi = \frac{\pi}{2}$ $\quad [S] = \begin{bmatrix} 0 & \alpha & j\beta & 0 \\ \alpha & 0 & 0 & j\beta \\ j\beta & 0 & 0 & \alpha \\ 0 & j\beta & \alpha & 0 \end{bmatrix}$

Antisymmetric: $\theta = 0, \varphi = \pi$ $\quad [S] = \begin{bmatrix} 0 & \alpha & \beta & 0 \\ \alpha & 0 & 0 & -\beta \\ \beta & 0 & 0 & \alpha \\ 0 & -\beta & \alpha & 0 \end{bmatrix}$

Directionnal coupler with Sonnet

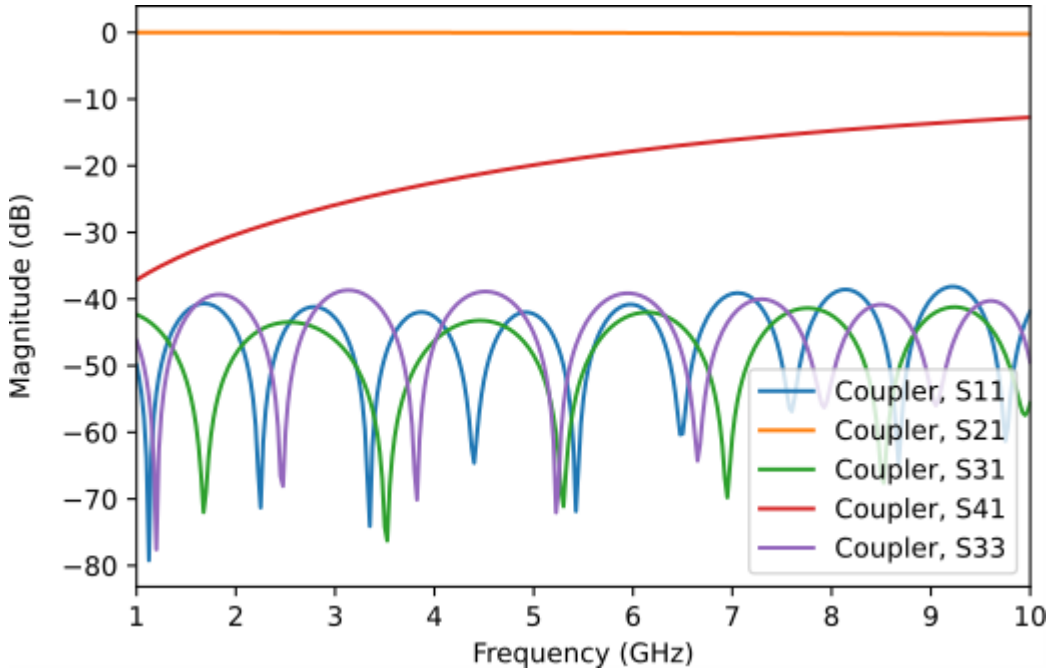


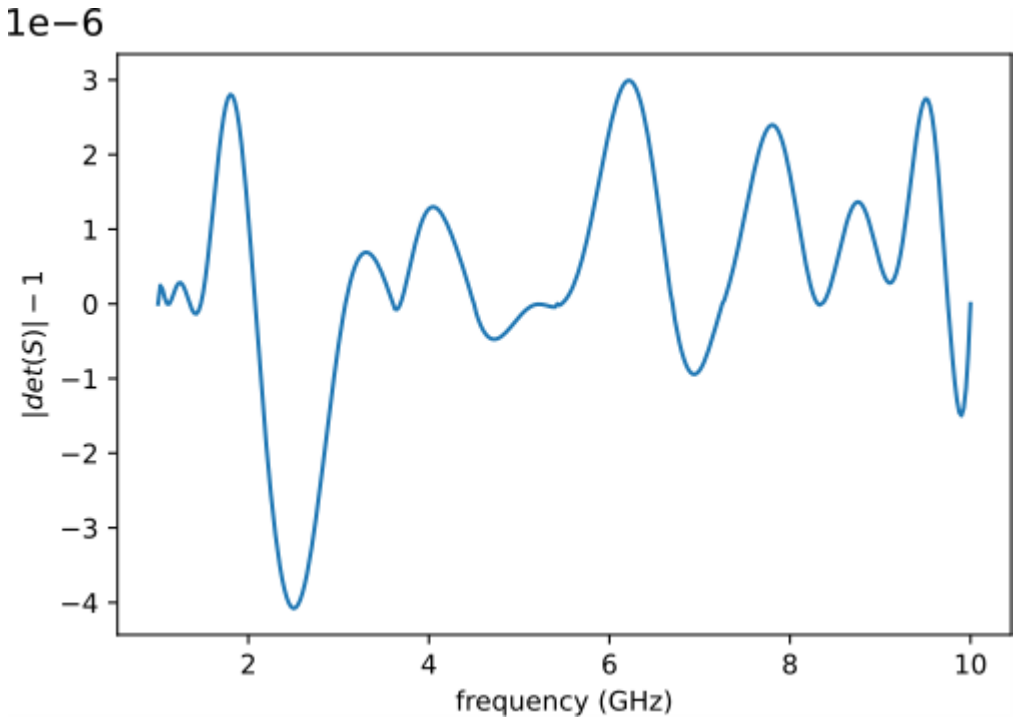
we adjuste the gap to get de coupling < -13 dB:

$S_{21} \sim -0.25\ dB$

$S_{41} < -12\ dB$

$S_{31} < -40\ dB$

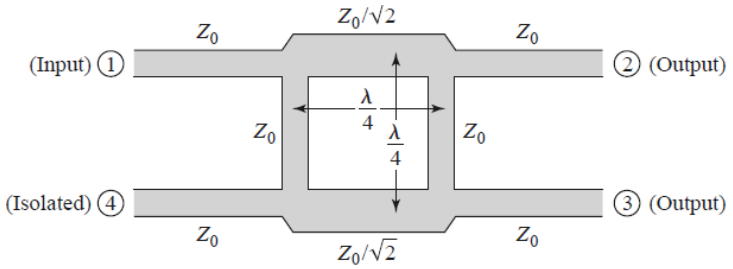




we get $|\det(S)| = 1$, if there is no dissipation (neither in substrate nor metal)

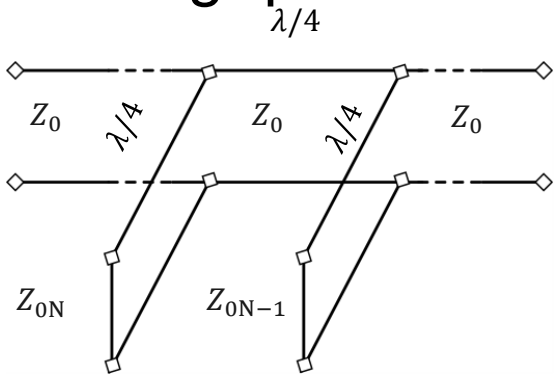
Branche-line 90° hybrid coupler

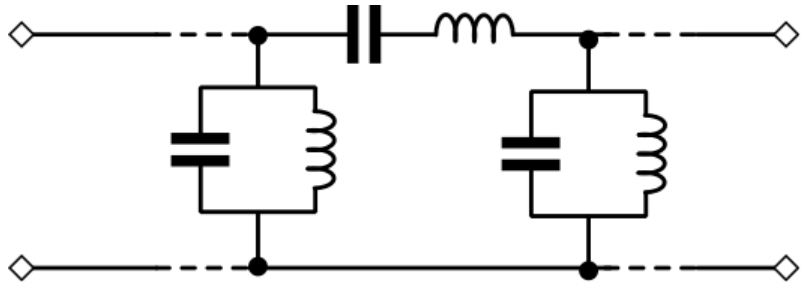
We want to build a µ-wave beam-splitter (divide the power)



Filters

Bandpass filters using quarter-wave resonators





for a fifth-order Chebyshev filter with 0.1 dB ripple, the normalized prototype values (g_i) are:

$$g_1 = g_5 = 1.1468, g_2 = g_4 = 1.3712 \text{ and } g_3 = 1.9750$$

then the characteritic impedance of the short-circuited $\lambda/4$ lines are $Z_n = \frac{2Z_0}{\pi g_n \text{ FBW}}$ and Finlay:

$$f_0 = 4 \text{ GHz}$$

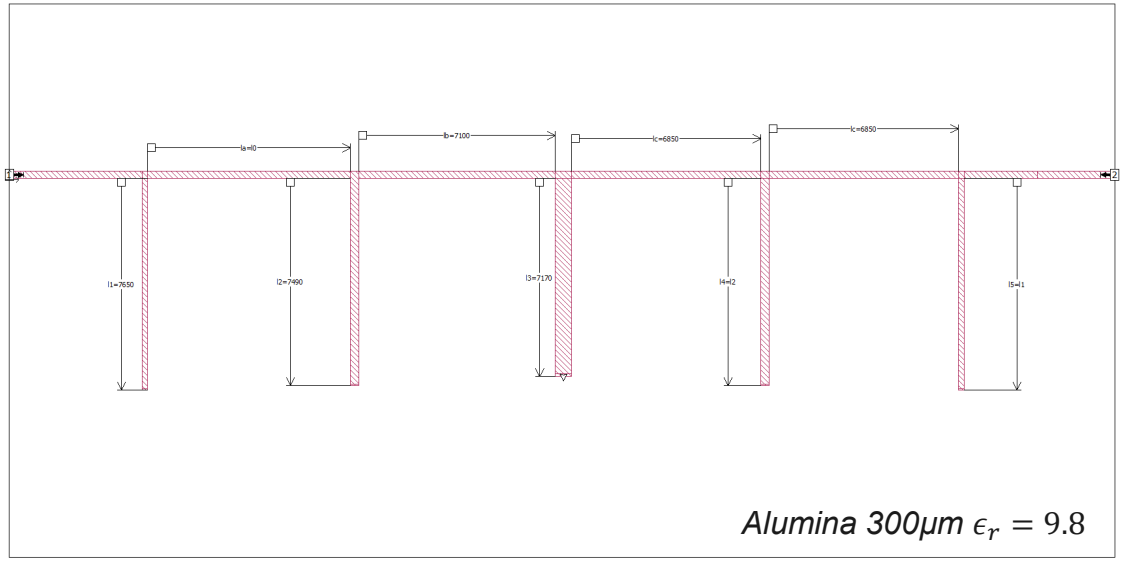
$$\Delta f = 2 \text{ GHz}$$

$$\text{FBW} = {}^{\Delta f}/_{f_0} = 0.5$$

n	g_n	$Z_n(\Omega)$	${}^{\lambda_n}/_{4}$ (mm)	$w_n(\mu\text{m})$
0		50	7.56	250
1,5	1.1468	55.5	7.59	200
2,4	1.3712	46.4	7.49	300
3	1.9750	32.6	7.17	600

Filters

Python script:



Alumina 300μm $\epsilon_r = 9.8$

$$f_0 = 4.08 \text{ GHz}$$

$$\Delta f = 3.12 \text{ GHz}$$